

Estimating Water Hammer Damping Ratios Using Continuous Wavelet Transform for Induced Hydraulic Fracture Complexity Characterization

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Summary

At the end of a hydraulic fracturing treatment, the pump shuts down, causing a water hammer (WH) effect to occur. The WH is influenced by formation properties and fracture complexity. In this study, we use the continuous wavelet transform (CWT) technique to analyze WHs, interpreting damping coefficients of the fracturing stages as indicators of induced fracture complexity within the same reservoir. We introduce a model that treats a WH as a damped harmonic oscillator, leveraging this analogy to understand WH signatures.

In this model, the WH signal is viewed as a damped harmonic oscillator. CWT is applied to the post-fracture wellhead signal across multiple scales, using the complex Morlet wavelet. Ridge detection traces dominant frequencies, forming a “ridge” that indicates the system’s natural frequency. Envelope calculation isolates this ridge, capturing amplitude and normalizing it to highlight decay characteristics. The damping coefficient is estimated from the linear regression envelope, where the slope represents the logarithmic decline.

In low-permeability reservoirs, fracture complexity and tortuosity strongly influence the damping of the WH signal. The methodology has been applied to various fracture stages, showing a strong correlation between damping coefficients and the number of propagating fractures. This technique automates the estimation of damping coefficients, linking fracture intensity and damping behavior, providing a valuable tool for evaluating induced fracture complexity.

Introduction

WH is an oscillatory pressure signal generated by an abrupt change in wellbore fluid velocity, such as when the pump rate is rapidly dropped to zero at the end of a fracturing treatment. After the fracturing pumps are shut down, a few minutes of pressure decline, including the WH signature, are routinely recorded during each fracture stage. Despite operators worldwide collecting extensive pressure decline data annually, these data are seldom utilized in post-stage analysis.

Holzhausen and Gooch (1985) were among the first to investigate the impact of hydraulic fracture growth on the free oscillations of wellbore pressure, providing valuable insights into fracture diagnostics. Their methodology involved modeling a fracture as a resistive-capacitive well termination to simulate field observations. This approach leveraged elasticity to define hydraulic resistance and capacitance based on fracture dimensions, thereby linking free oscillations to fracture geometry. Two field experiments supported their model. In the first, a well cased to 1589 m with perforations in a low-permeability production zone exhibited a doubling of oscillation periods post-fracturing. The second experiment in Glenpool, Oklahoma, USA, demonstrated that larger water injection volumes into fractured sandstone formations resulted in prolonged pressure oscillations. Their findings revealed that hydraulic fracture growth significantly influences the amplitude and frequency of free oscillations, and their model provided a means to estimate fracture dimensions from oscillation frequency and attenuation.

Soliman and Azari (1998) further advanced this field by delving into the significant roles of friction and fluid leakoff in hydraulic fracturing operations. His methodology encompassed field experiments where fluid was injected into wells, and the resultant pressure responses were meticulously recorded. The authors developed sophisticated mathematical models that incorporated the dynamics of friction and leakoff to analyze their effects on fracture parameters, including length, width, and conductivity. The study revealed that friction and leakoff substantially influence the pressure profiles observed during hydraulic fracturing. Ignoring these factors could result in significant inaccuracies in estimating the critical fracture parameters. Soliman and Azari concluded that for hydraulic fracturing models to be accurate and effective, they must integrate the effects of friction and leakoff, leading to better fracture design, reduced operational costs, and improved overall efficiency.

Mondal (2010) provided a comprehensive methodology for using WH effects in fracture diagnostics. By developing a semi-analytical model to simulate WH phenomena, Mondal offered valuable insights into wellbore stability and fracture dimensions, contributing to improved completion design and sand control in injectors.

Carey et al. (2015) contributed to the understanding of WH properties for wellbore connections by developing a numerical model to simulate WH signatures. They represented the wellbore with hydraulic fractures as a circuit comprising resistance (R), capacitance (C), and inductance (I) connected in series. By iteratively adjusting the values of R, C, and I, they achieved a match between field data and simulated results. Subsequently, they performed a sensitivity analysis to assess the impact of each parameter on the WH signal behavior. Their analysis concluded that WH pressure signatures are significantly influenced by the created fractures.

Hwang et al. (2017) presented a comprehensive analysis of using WH signatures as a diagnostic tool for evaluating fracture dimensions and stress interference in multistage hydraulic fracturing treatments. The authors developed a simulation model that treats the wellbore fracture system as an R-C-I circuit, allowing for the interpretation of WH data to estimate fracture geometries. Their study demonstrates that variations in the WH signature, influenced by factors such as fracture compliance and near-wellbore friction, can provide valuable insights into fracture dimensions, including height, length, and width. The authors also address the interstage stress shadow effect,

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proposing a method to estimate instantaneous shut-in pressure (ISIP) and minimum horizontal stress (S_{hmin}) changes over multiple stages. Their findings suggest that WH analysis may complement traditional diagnostic methods, such as microseismic monitoring, providing a cost-effective means of hydraulic fracture evaluation and enhancing the accuracy of stress interference analysis.

Iriarte et al. (2017) analyzed a data set of more than 100 wells across three different US plays. They characterized WH signatures by calculating the decay rate, amplitude, duration, and period. To determine the decay rate, they fitted an exponential decay function to the WH peaks. Once these parameters were determined, they evaluated the effects of the wellbore, stimulation, and formation by correlating WH parameters with the parameters of interest for each segment. Their findings indicated that the amplitude of the WH is closely correlated with the number of perforations per stage, pump rate, and friction. In contrast, the duration and decay rate were found to correlate with fluid type, fluid volume, and the complexity of the fracture network.

Roussel et al. (2021) investigated the diagnostic value of ISIP in hydraulic fracturing operations. Using a novel geophysical signal processing-inspired technique, they automated ISIP calculation and demonstrated its significance in understanding local reservoir stresses and hydraulic fracture properties. Their work emphasized the importance of routine ISIP data collection for optimizing fracturing operations.

Sheludko et al. (2024) explored the use of machine learning to analyze WH signatures, aiming to classify fracture stage contributions to a well's production profile. Their study demonstrated that WH characteristics, such as initial amplitude, decay rate, phase angle, and angular frequency, could effectively indicate hydraulic fracture and reservoir properties. The authors developed a Python algorithm to automatically identify surface pressure and fluid pump rate channels, detect end-of-pumping events, and parse out WH and pressure decline signals from raw data. Using these characteristics, they trained a random forest classification model, achieving a test classification accuracy of 0.71 and an F1 score of 0.72. This model is used to estimate the relative productivity of hydraulic fracture stages.

Garbino et al. (2024) introduced an automated workflow for estimating ISIP in hydraulic fracturing operations, addressing the challenge of pressure oscillations, known as WH, that complicate ISIP interpretation. The proposed method integrates signal processing and an optimization algorithm, offering improved consistency and accuracy compared with traditional manual interpretations. Field validation demonstrated the workflow's robustness in handling varying shut-in conditions and reliably estimating ISIP. Additionally, the study investigated the potential of using WH frequency as a diagnostic tool for stage isolation, although the results were inconclusive.

Wavelet analysis has driven significant advancements in the oil industry, beginning with Jansen and Kelkar (1997), who utilized wavelet transformation to analyze nonlinear, nonstationary interwell relationships by decomposing production data into frequency and smoothed components, enabling robust, time-dependent cross-correlation, such as Spearman rank correlation. Soliman et al. (2003) further pioneered its use in well test analysis and fracturing applications, followed by Guan et al. (2004), who provided a comprehensive review emphasizing its effectiveness in extracting critical insights from complex signal data. Chandrasekhar and Eswara Rao (2012) applied wavelet analysis to well log data, successfully identifying productive pay zones. Leveraged wavelet analysis to interpret fracturing data, enabling the detection of hydraulic fracturing issues and anomalies in fracture behavior, while Unal et al. (2019) and Eltaleb et al. (2020) expanded its application to identify closure pressures in diagnostic fracture injection tests. Unal et al. (2019) introduced a method for estimating interwell connectivity using wavelet transforms of injection and production rate data, utilizing the variance of detail coefficients and cross-correlation techniques. Awad et al. (2020) and Gabry et al. (2023a, 2023b, 2024a, 2024b) applied wavelet analysis to detect fracture propagation events and estimate microseismic activity associated with hydraulic fracturing. These collective efforts underscore the transformative potential of wavelet analysis in addressing complex challenges within the oil and gas industry, demonstrating its versatility across well test analysis, reservoir management, and hydraulic fracturing optimization.

As demonstrated in previous studies, modeling WH as a damped harmonic oscillator relies on parameter optimization techniques, often resulting in nonunique solutions that limit real field application. In our study, we introduce a novel WH modeling and diagnostic approach, also treating the phenomenon as a damped harmonic oscillator, building on the models proposed by Roussel et al. (2021), Sheludko et al. (2024), and Garbino et al. (2024). While these studies utilized linear parameter optimization in their oscillator models, our approach enhances the methodology by incorporating a new equation for the oscillator and using the CWT with a complex Morlet wavelet to accurately calculate the damping coefficient. This refinement offers a more precise representation of pump shutdown pressure responses across various WH signal scenarios. The calculated damping coefficient, in particular, serves as a reliable indicator of induced fracture complexity, effectively capturing the interaction between the wellbore and the formation. Following the determination of the damping coefficient, we apply nonlinear optimization to fit the remaining parameters, achieving a more consistent solution and significantly improving the accuracy of fracture complexity assessments.

Induced fracture complexity refers to the intricate network of fractures created during hydraulic fracturing, which plays a critical role in enhancing fluid flow and maximizing reservoir contact. Understanding and accurately measuring fracture complexity is essential for optimizing stimulation treatments. Induced fracture complexity is significantly influenced by the choice of fracturing fluids. Carey et al. (2015) found that crosslinked gel treatments tend to create wider fractures with higher decay rates due to signal dampening, while slick-water treatments often produce longer fractures with lower decay rates, supporting more planar fracture propagation. Ciezobka et al. (2016) also observed that higher decay rates correlate with more complex fracture networks near the wellbore as seen in **Fig. 1**. In multistage hydraulic fracturing treatments with similar permeability ranges and consistent fracturing designs, our technique's damping coefficient serves as a reliable indicator of fracture complexity, offering valuable insights for optimizing fracturing operations.

WH modeling has been widely used in various applications, such as automated workflows for estimating ISIP, as demonstrated by Roussel et al. (2021) and Garbino et al. (2024), and for classifying stage contributions to well production, as shown by Sheludko et al. (2024). Additionally, it has been used for evaluating fracture stage efficiency (Iriarte et al. 2017) and fracture diagnostics in multistage hydraulic fracturing to assess fracture dimensions and stress interference (Holzhausen and Gooch 1985; Soliman and Azari 1998; Mondal 2010; Carey et al. 2015; Hwang et al. 2017). These studies collectively underscore the importance of analyzing WH signals and other pressure transients in hydraulic fracturing. By leveraging advanced modeling techniques, signal processing methods, and machine learning, researchers continue to enhance our understanding of fracture networks, fracture stage productivity, and reservoir properties.

Our novel approach enhances these models by delivering a more robust and detailed analysis of fracture complexity, utilizing wavelet-based damping coefficients derived through the CWT with complex Morlet wavelets—the same wavelet method used by Gabry et al. (2023a, 2023b, 2024a, 2024b). This technique has been validated against fracture identification logs, demonstrating high accuracy and reliability. This innovative method offers a cost-effective solution for evaluating fracture complexity at each hydraulic fracture stage, optimizing both stage productivity and overall fracturing operations.

Methodology

Our methodology presents an enhanced model of WH, conceptualized as a damped harmonic oscillator combined with an underlying slow pressure decay, similar to the models proposed by Roussel et al. (2021), Sheludko et al. (2024), and Garbino et al. (2024) but with key

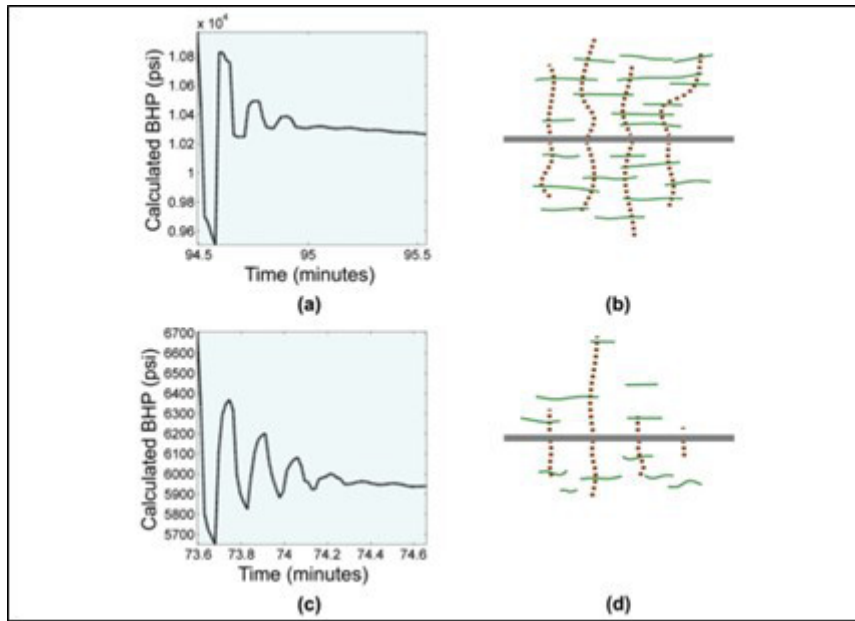


Fig. 1—Typical WH response with (a) high decay rates and (c) low decay rates as well as possible reasons based on our hypotheses (b) and (d) (Ciezobka et al. 2016). BHP = bottomhole pressure.

modifications. We apply the CWT using a complex Morlet wavelet for automated oscillatory analysis to accurately calculate the damping coefficient, followed by nonlinear parameter optimization techniques to fit the remaining parameters.

WH as Damped Harmonic Oscillator. In our approach, we introduce the WH as a damped harmonic oscillator (P_{DHO}) with exponential decay (P_{ED}), similar to the models presented by Roussel et al. (2021) and Sheludko et al. (2024). While their models are used to detect the ISIP, our model focuses on detecting induced fracture complexity. To understand the WH phenomenon, it is essential to emphasize its physical concept. WH occurs when pressure waves travel back and forth through the pipe and fracture at sonic velocities. The gradual decrease in pressure amplitude follows an exponential pattern, attributed to the slow seepage of fluid into the surrounding formation and its fractures.

The WH phenomenon, which describes transient pressure oscillations in hydraulic systems caused by sudden flow changes, can be modeled by a partial differential equation derived from the conservation of mass and momentum. Assuming a compressible fluid and small perturbations in pressure and velocity, the governing partial differential equation for pressure $p(x, t)$ in space and time is expressed as in Eq. 1:

$$\frac{\partial^2 p}{\partial t^2} = \frac{K}{\rho_o} \frac{\partial^2 p}{\partial x^2} - \gamma \frac{\partial p}{\partial t}, \quad (1)$$

where K is the bulk modulus of the fluid and ρ_o is the reference density. The damping coefficient γ accounts for frictional losses in the system, which are proportional to the velocity. This partial differential equation captures the wave-like propagation of pressure changes, including the effects of damping over time. We consider a system with boundary and initial conditions representing a sudden pressure change, such as a valve closure. To solve this equation, assume a separable solution of the form $P(x, t) = X(x)T(t)$. Substituting this into the partial differential equation, then dividing through by $X(x)T(t)$ and rearranging terms yields Eq. 2:

$$\frac{1}{T(t)} \frac{\partial^2 T(t)}{\partial t^2} + \frac{\gamma}{T(t)} \frac{\partial T(t)}{\partial t} = \frac{K}{\rho_o} \frac{1}{X(x)} \frac{\partial^2 X(x)}{\partial x^2}. \quad (2)$$

Since the left-hand side depends only on t and the right-hand side depends only on x , both must equal a constant, $-\lambda^2$, leading to two ordinary differential equations—one for time and one for space. In the context of a hydraulic fracturing wellbore, pressure is measured at the wellhead, eliminating the need for the spatial equation. Consequently, the time-dependent equation becomes Eq. 3, with $T(t)$ replaced by $P_{\text{DHO}}(t)$ to reflect the focus on pressure after removing the spatial dependency:

$$\frac{\partial^2 P_{\text{DHO}}(t)}{\partial t^2} = -\gamma \frac{\partial P_{\text{DHO}}(t)}{\partial t} - \omega^2 P_{\text{DHO}}(t). \quad (3)$$

Eq. 3 describes the pressure change over time, incorporating the effects of damping, represented by $(-\gamma \frac{\partial P_{\text{DHO}}(t)}{\partial t})$ and an oscillatory term, $\omega^2 P$. In the context of WH, pressure oscillates like a wave while losing energy through exponential decay. We propose a solution in the form of Eq. 4. Substituting this proposed solution into Eq. 3 verifies that it satisfies the equation perfectly, confirming that the guessed solution is valid.

$$P_{\text{DHO}}(t) = e^{-\gamma t} \sin(2\pi f t + \theta). \quad (4)$$

In some systems, such as hydraulic fracturing, pressure also decays due to fluid leakoff and other dissipative effects that do not contribute to oscillations. To account for these effects, an additional exponentially decaying term and a constant are included, as shown in Eq. 5:

$$P_{ED} = be^{at} + c. \quad (5)$$

By combining these two effects on the pressure wave, it is possible to simulate the WH pressure wave, as depicted in Eq. 6. This approach allows us to capture the oscillatory nature of the pressure waves and their exponential decay, providing a comprehensive understanding of the induced fracture complexity.

$$P = P_{DHO} + P_{ED} = Me^{-\gamma t} \sin(2\pi ft + \theta) + be^{-at} + c, \quad (6)$$

where M represents the magnitude of the WH, which is the amplitude of the oscillation; γ is the damping coefficient; f is the frequency of the oscillation; and θ is the phase shift. Constants a , b , and c account for additional exponential decay and offset components.

The exponential decay factors a and b primarily depend on the leakoff parameter of the formation. The parameter c depends on the minimum horizontal stress of the formation and the net pressure during the fracture stage. We use a sine wave function because, during a WH event, the pressure initially rises due to the shutdown of the fracture pump and then oscillates back and forth. **Figs. 2 and 3** illustrate the impact of varying each parameter in the model, including amplitude M , frequency f , damping coefficient γ , phase shift θ , and the exponential decay factors a , b , and c , on the WH pressure oscillations. These figures reveal that similar WH signatures can be produced by different combinations of parameters. This observation underscores the importance of accurately quantifying the damping coefficient using CWT. By estimating the damping coefficient first, we can fine-tune the other parameters to achieve a unique match with the actual field data.

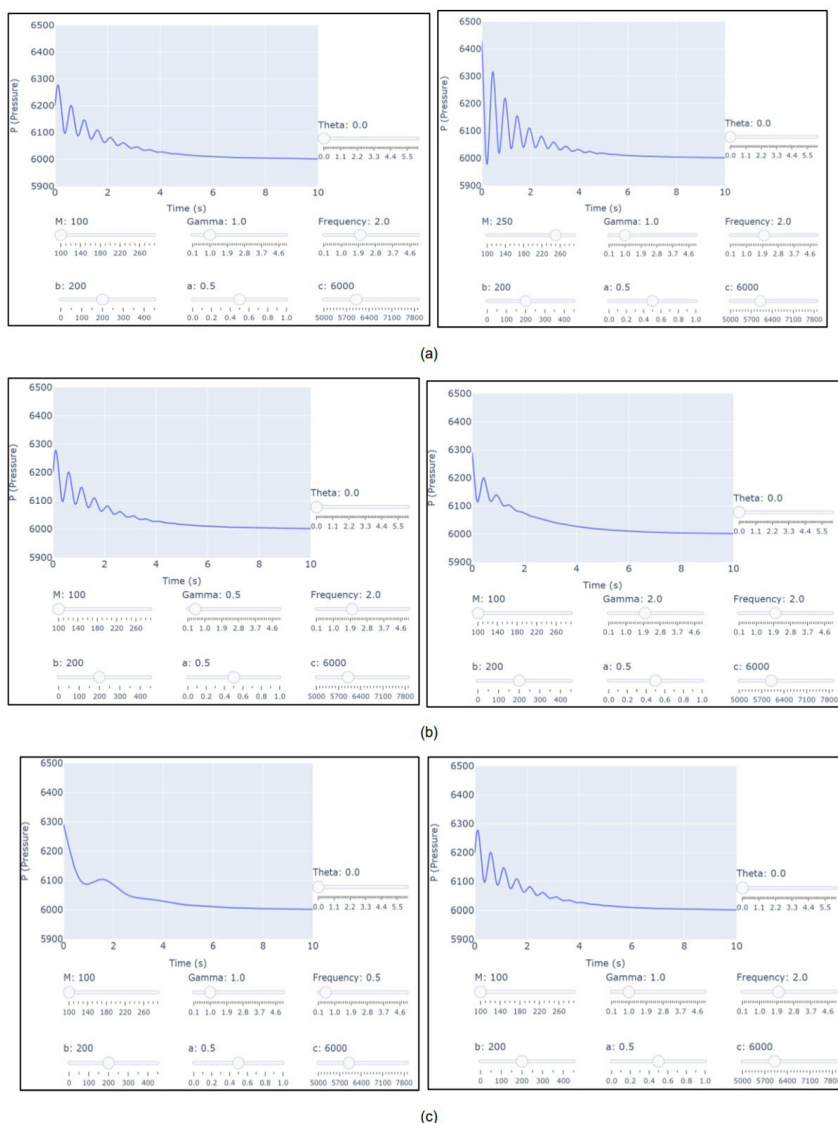


Fig. 2—Effect of changing the damping coefficients on the damped harmonic oscillator: (a) changing M , (b) changing γ , and (c) changing f .

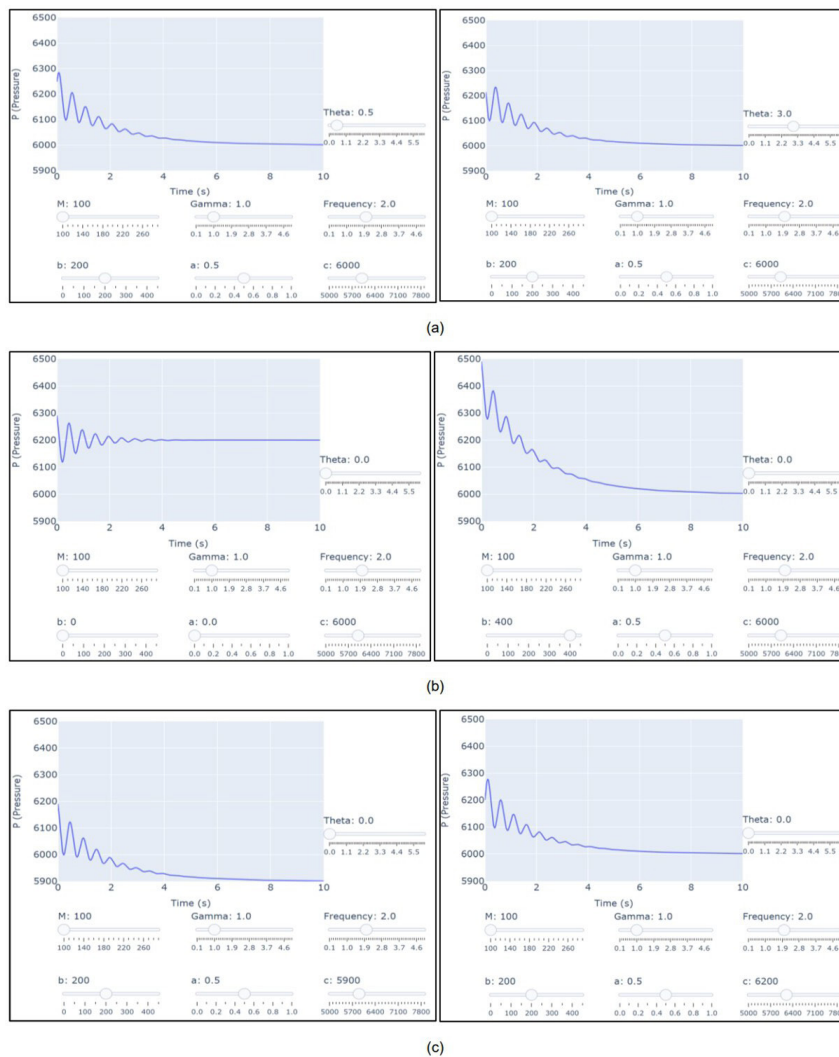


Fig. 3—Effect of changing the damping coefficients on the damped harmonic oscillator: (a) changing θ , (b) changing a and b , and (c) changing c .

In this model, the amplitude M represents the magnitude of the WH, corresponding to the maximum deviation from the steady-state pressure, which is influenced by the initial momentum change in the fluid, such as from rapid valve closures. It controls how much the pressure increases when the system is disturbed. The damping coefficient γ corresponds to the rate at which the oscillations diminish, which is affected by factors such as friction, viscosity, pipe material, diameter, and fluid properties, including the presence of gas. It controls how quickly the oscillations diminish due to system complexity, controlling the number of pressure wave oscillations. The frequency f of the oscillation, represented by $2\pi f$ in the equation $\sin(2\pi f t + \theta)$, dictates the rate of pressure wave oscillations within the pipe, determined by the speed of sound in the fluid and the pipe's characteristics, such as length and elasticity. The phase shift θ influences the initial position of the pressure wave oscillation, affecting whether the pressure initially spikes or is controlled by leakoff. The constants a and b represent the late-stage leakoff in the fracture geometry and formation, affecting the exponential decay of pressure.

The damping of a harmonic oscillator is a crucial aspect that describes how the oscillations decrease over time due to energy dissipation in the system. In a damped harmonic oscillator, the damping force acts opposite to the direction of motion and is proportional to the velocity of the oscillation. This damping force can arise from various sources, such as friction, air resistance, or internal material resistance.

The damping coefficient, denoted by γ , quantifies the extent of damping. Depending on its value, the system can be categorized into underdamped, critically damped, or overdamped regimes as seen in **Fig. 4**. In an underdamped system ($\gamma < \gamma_{\text{critical}}$), the oscillator continues to oscillate with gradually decreasing amplitude. In a critically damped system ($\gamma = \gamma_{\text{critical}}$), the system returns to equilibrium as quickly as possible without oscillating. In an overdamped system ($\gamma > \gamma_{\text{critical}}$), the system returns to equilibrium without oscillating but more slowly than in the critically damped case.

CWT. The CWT is utilized to accurately determine the damping coefficient in WH modeling. It acts as a flexible mathematical microscope, effectively capturing the oscillatory behavior of the WH. This approach enables precise estimation of the damping coefficient, providing a unique solution for the damped harmonic oscillator model of WH, as opposed to the parameter-matching methods used in previous studies.

Wavelet transforms, first introduced by Haar (1910), have revolutionized signal processing by offering significant advantages over traditional methods such as Fourier analysis. Unlike Fourier transforms, which excel at analyzing stationary signals but struggle with time-varying (nonstationary) signals, wavelet analysis provides detailed insights into trends, discontinuities, and higher-order derivatives

Water Hammer Signature with Different Damping Ratios

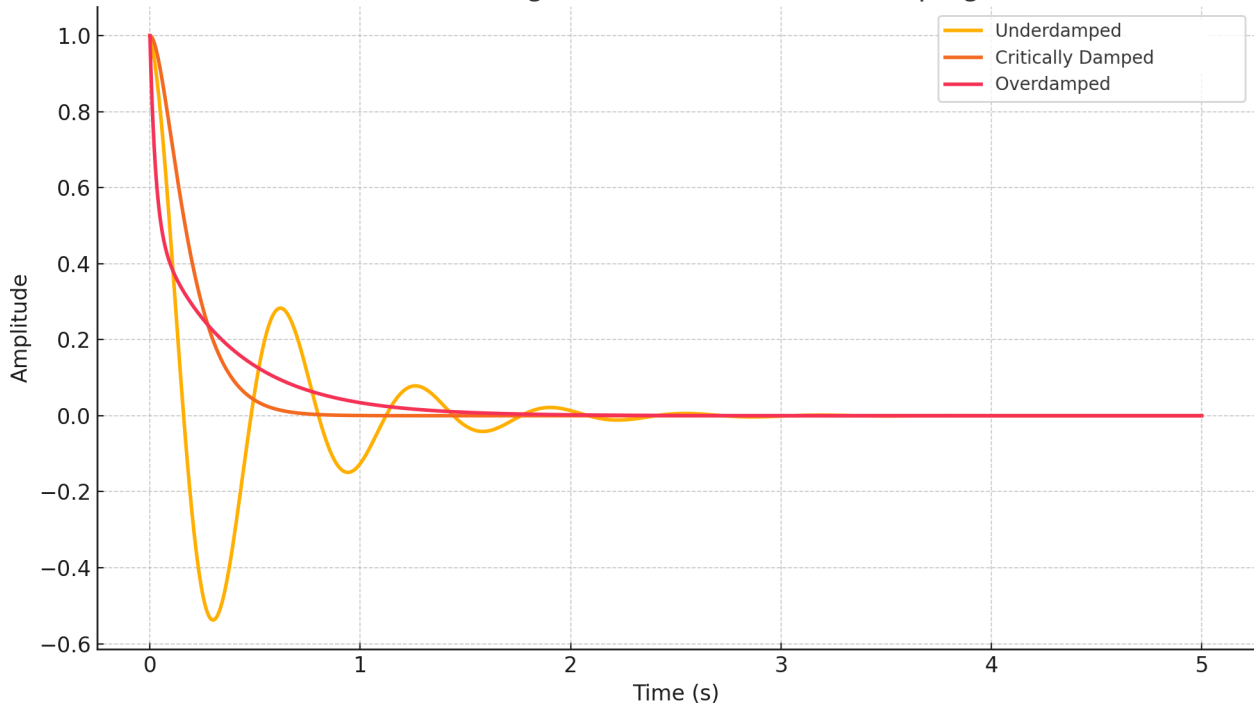


Fig. 4—Underdamped, critically damped, or overdamped harmonic oscillators.

(Rioul and Vetterli 1991). This makes wavelet transforms indispensable across a wide range of applications, including cardiac monitoring (Adeli et al. 2003), kidney performance, climate studies, and wellsite data interpretation.

Fourier transforms are widely used in signal processing to provide frequency information, converting time-domain signals into combinations of sine and cosine waves. However, they fail to indicate when specific frequency components occur, making them less suitable for nonstationary signals. The windowed Fourier transform sought to address this limitation by dividing the signal into smaller stationary segments using window functions. However, it remains prone to aliasing when high- and low-frequency components extend beyond the window's range, resulting in potential inaccuracies.

Wavelet transforms overcome these challenges with their ability to represent signals in both time and frequency domains simultaneously. This dual-domain capability makes them particularly effective for analyzing signals with a wide range of dominant frequencies, providing high-resolution time-frequency localization. Wavelets are pulse-like in shape and offer a flexible “mathematical microscope,” capable of magnifying small details and uncovering hidden features in complex signal data. This versatility is especially valuable in well-site applications, where parameters such as pressure require accurate interpretation.

A key variant of wavelet analysis, the CWT, enhances this capability by not requiring a uniform sampling rate and effectively capturing all signal details. CWT, especially when paired with complex wavelets like the Morlet wavelet, allows for the detection of fractal dimensions, which are critical for characterizing complex systems. This approach has been widely adopted in well test analysis and hydraulic fracturing diagnostics, as demonstrated by Gabry et al. (2023a, 2023b) and Gabry et al. (2024a, 2024b). The complex Morlet wavelet is particularly favored for its ability to deliver detailed signal analyses across various engineering applications. Notably, the choice of wavelet type often has minimal impact on results as long as it aligns with the signal's characteristics.

The wavelet transform process involves two main steps—displacement across the signal (translation) and compression/expansion (scaling). It is essentially a convolution operation between a small wavelet function and the signal, allowing for localized matching. This matching is characterized by the scale (dilation parameter, a) and location (translation parameter, b) of the convolution in the time domain.

There are two main types of wavelet transforms—discrete wavelet transform and CWT. CWT, widely adopted since the 1990s, involves a continuous convolution process, where a wavelet is applied to the acquired signal. It achieves a detailed signal representation by dynamically adjusting wavelet translation and scale to effectively highlight subtle variations. Mathematically, the wavelet transform is expressed as Eq. 7.

$$T(a, b) = w(a) \int_{-\infty}^{\infty} x(t) \psi^* \left(\frac{t-b}{a} \right) dt, \quad (7)$$

where $w(a)$ is a weighting factor (typically $\frac{1}{\sqrt{a}}$ for energy conservation), $x(t)$ represents the signal, and ψ is the mother wavelet. For complex wavelets, ψ becomes ψ^* , indicating the use of the complex conjugate of the wavelet function. The wavelet transform is performed by convolving the mother wavelet with the signal. The wavelet is then translated over the signal to calculate the wavelet transform coefficients at different positions b . Subsequently, the mother wavelet is scaled by different factors a , and the process is repeated to compute the coefficients at varying scales and positions. This process allows for the analysis of the signal's features across different scales and locations, as illustrated in Fig. 5. Wavelets come in a variety of shapes and can be either real (e.g., Coiflets, Daubechies, Symlets, and Haar) or complex (e.g., the complex Mexican Hat wavelet, Morlet wavelet, complex Ricker wavelet, complex Haar wavelet, and complex Shannon wavelet). The diverse forms of wavelets enable their application across a wide range of fields, including medical and electrical engineering, as seen in Fig. 6. When a complex wavelet is used, the modulus of the complex wavelet coefficient is used for analysis and

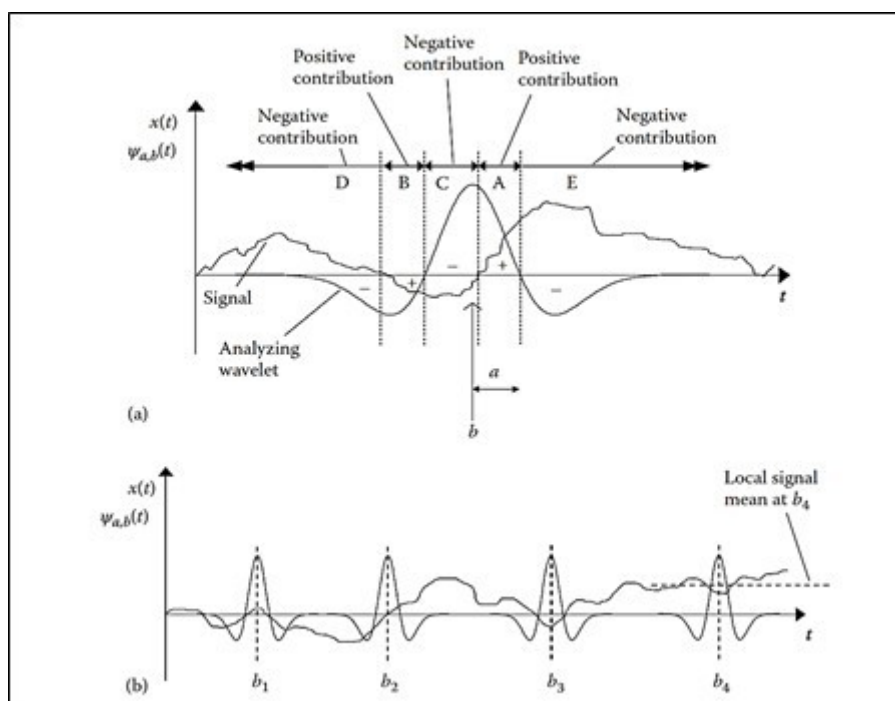


Fig. 5—Wavelet transform of the signal: (a) wavelet at specific dilation and location, with regions of positive and negative contributions marked, and (b) fixed dilation wavelet at four distinct locations, showing varying transform values at each point (Addison 2017).

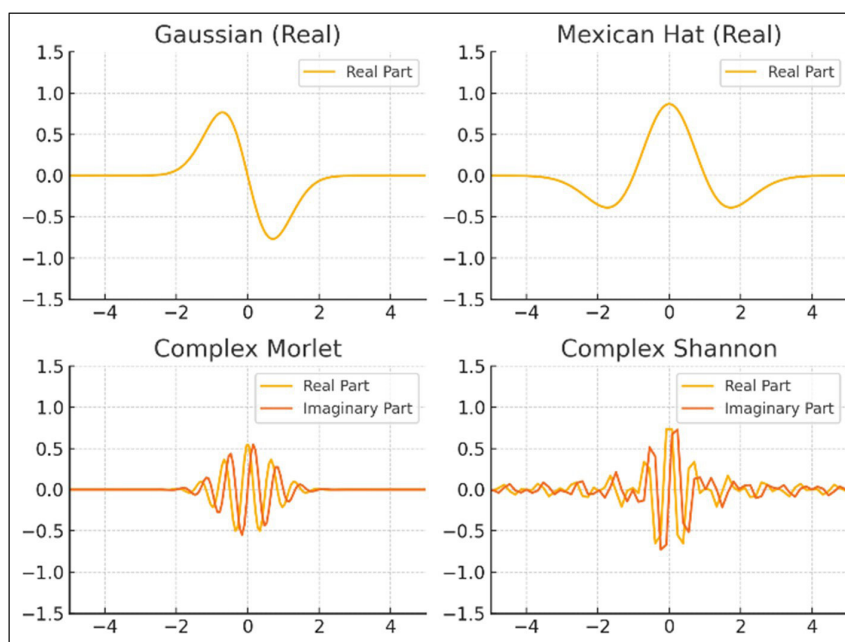


Fig. 6—Examples of wavelets used for CWT.

visualization. This modulus, referred to as the wavelet transform modulus, provides a comprehensive representation of the signal by combining information about its amplitude and phase.

In practical applications, such as analyzing pressure and temperature data at wellsites, CWT serves as a mathematical microscope, discerning complex signals and revealing information that might be obscured in traditional time-amplitude plots or Fourier transforms. This makes wavelet transforms a powerful tool for signal analysis, offering insights into signal characteristics that might otherwise remain hidden and significantly enhancing our ability to interpret and analyze complex data across various scientific and engineering disciplines. In our approach, we use complex Morelet wavelets for our CWT. The Morlet wavelet, characterized by a Gaussian-windowed sinusoid, is particularly well-suited for analyzing oscillatory signals because it provides a good balance between time and frequency resolution. The complex nature of the Morlet wavelet also allows for the capture of both amplitude and phase information, which is crucial for detailed signal analysis. The complex Morlet wavelets can be seen in Eq. 8:

$$\psi^*(t) = \frac{1}{\sqrt{\pi B}} e^{\frac{t^2}{B}} e^{i2\pi Ct}, \quad (8)$$

where B is the bandwidth and C is the center frequency. In our approach, a fixed bandwidth ($B = 1.5$) and center frequency ($C = 1$) have been empirically determined for optimal performance. The wavelet transform converts the signal to a magnified version of the signal called, as seen in **Fig. 7**. The wavelet transform provides a powerful tool for detecting the embedded features in the WH signal as it successfully captures the oscillatory behavior and energy loss regime embedded in the signal.

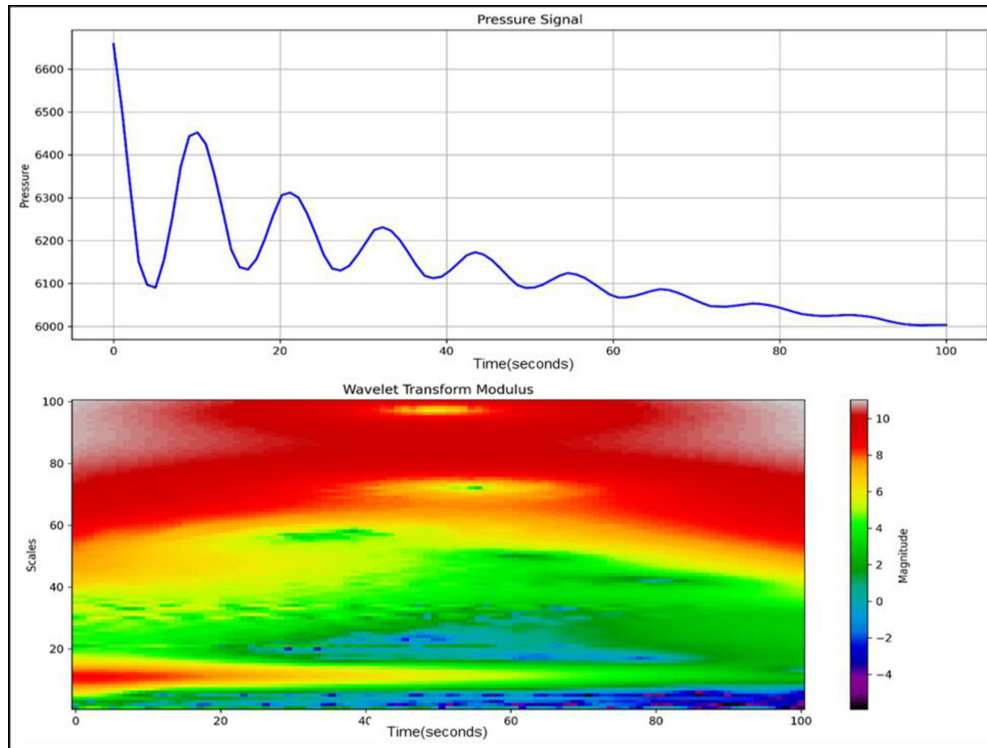


Fig. 7—Wavelet transform modulus of an example of WH pressure wave.

WH Damping Estimation. The damping coefficient quantifies the rate at which oscillations in a system decay following a disturbance. It is a critical parameter in analyzing real field data across various domains, including mechanical engineering, structural dynamics, and signal processing, as demonstrated by Slavič et al. (2003), Chen et al. (2009), and Mihalec et al. (2016). These studies automated the identification of the damping coefficient in dynamic mechanical systems. A similar approach is used here to analyze WH signals, providing valuable insights into their oscillatory behavior and decay characteristics. To estimate the damping coefficient using the CWT, we start with the pressure signal of interest and its sampling rate. The signal length is selected to cover the period from the end of oscillations caused by pump shutdown to the onset of formation leakoff dominance. The next step is to define a range of scales for the wavelet transform, which allows for analyzing the signal at multiple resolutions.

The wavelet transform is applied using the complex Morlet wavelet, which consists of a complex exponential function modulated by a Gaussian window. This wavelet is ideal for capturing both the frequency and phase characteristics of the signal due to its balance between time and frequency localization. For this study, a bandwidth parameter of 1.5 and a center frequency of 1 are selected, as they provide effective resolution for analyzing transient and oscillatory behavior in WH signals.

Using these predefined scales, the CWT of the signal is computed, resulting in a coefficient matrix and corresponding frequencies ($\frac{1}{\text{scale}}$). This matrix represents the wavelet coefficients at different scales and time points, revealing how the signal's frequency content evolves over time. The ridge of the wavelet transform is identified by finding the indices of the maximum absolute values of the wavelet coefficients along the scale axis as shown in Eq. 9.

$$a_{\text{ridge}} = \operatorname{argmax}_a |T(a, b)|, \quad (9)$$

where $T(a, b)$ along this ridge represents the signal's skeleton, which highlights the dominant oscillatory behavior. These indices correspond to the points of highest energy at each timestep, isolating the dominant frequency components of the signal. The frequencies associated with these maximum coefficients define the ridge of the wavelet transform. By extracting the wavelet coefficients along this ridge, a skeleton of the signal is constructed—a simplified representation that retains its essential oscillatory behavior. The wavelet transform isolates the oscillatory component of the model, effectively separating it from the pressure's exponential decay caused by formation leakoff. To analyze the signal's amplitude modulation, the envelope of the skeleton is computed. This envelope provides a smooth curve that outlines the upper bounds of the oscillations, capturing amplitude variations over time.

To estimate the damping coefficient, a logarithmic transformation is applied to the envelope. This transformation linearizes the exponential decay characteristic of damped oscillatory signals, enabling the quantification of damping behavior. A linear slope is then fitted to the logarithm of the envelope using the least squares method, with the fitting time window determined by the length of the envelope.

Mathematically, the envelope $E(t)$ of the skeleton is calculated using the modulus of the wavelet coefficients, as described in Eqs. 10 and 11. This approach leverages the property that the signal's envelope decays exponentially over time, providing a straightforward method for analyzing damping characteristics.

$$E(t) = |T(a, b)|, \quad (10)$$

$$\ln E(t) = \ln E_0 - \gamma t, \quad (11)$$

where E_0 is the quantity $M \sin(2\pi ft + \theta)$, t is the time, and γ is the damping coefficient.

In summary, this methodology involves several key steps as seen in **Fig. 8** including defining a range of scales for the wavelet transform, performing the CWT using the complex Morlet wavelet, identifying the ridge and skeleton of the transform, computing the envelope of the skeleton, fitting a linear slope to the logarithm of the envelope, and deriving the damping coefficient from the slope of the fit. The use of the complex Morlet wavelet, with its ability to capture both frequency and phase information, is crucial for accurately analyzing and estimating the damping behavior of oscillatory signals.

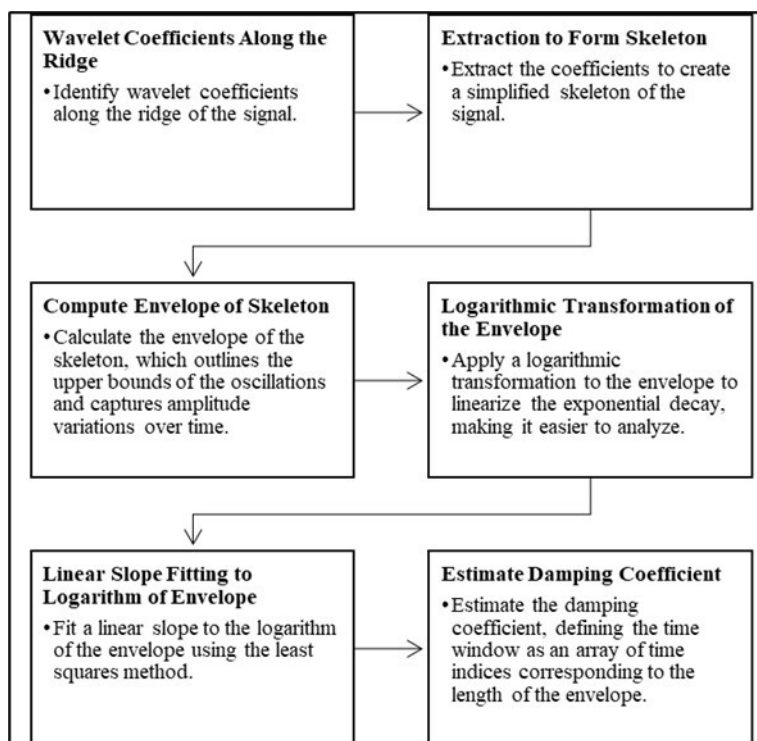


Fig. 8—Workflow of damping coefficient estimation using CWT.

Damping Estimation for Induced Fracture Complexity. Hydraulic fractures can be artificially divided into four categories: simple fractures, complex fractures, complex fractures with fissure openings, and complex fracture networks, as illustrated in **Fig. 9** (Warpinski et al. 2009). Fracture complexity during hydraulic fracturing is primarily determined by reservoir properties, including the pre-existing rock fabric and stress regime. Reservoirs with a high density of natural fractures or fissures are more prone to developing complex fractures. The stress regime is particularly crucial; the potential for complex fracture growth increases as the difference between minimum and maximum horizontal stresses decreases.

While operational techniques can influence fracture complexity, it is predominantly governed by these intrinsic reservoir characteristics. Fluid viscosity is a key operational factor: High-viscosity fluids tend to reduce fracture complexity, whereas low-viscosity fluids promote it. The decision to control or enhance fracture complexity depends on factors such as reservoir permeability, proppant distribution, and the desired fracture conductivity.

Estimating the induced fracture complexity is crucial for predicting post-fracture production. The distribution of proppant within complex fractures greatly impacts the conductivity requirements. In scenarios of network fracture growth, if injected proppant is concentrated in a single planar fracture with infinite conductivity, the network's overall conductivity requirements can be significantly reduced. This suggests that the conductivity of unpropped fractures can play a vital role in well productivity, particularly in water fractures and low-viscosity hybrid fracture treatments as discussed by Cipolla et al. (2010). The necessary fracture conductivity for optimal production varies based on factors such as reservoir permeability, fracture complexity, fracture spacing, and network width. Notably, these conductivity requirements do not always conform to simple dimensionless conductivity design guidelines. In network fracture growth cases, conductivity needs are proportional to the square root of fracture spacing, indicating that less conductivity is required as fracture spacing decreases. In low-permeability reservoirs, exploiting fracture complexity with an infinite conductivity primary fracture may be feasible if the unpropped fracture conductivity is sufficient. However, if uniform conductivity is required throughout the network, significantly higher conductivity levels are needed to achieve optimal production. This suggests that fully leveraging fracture complexity may be challenging, except in very low-permeability reservoirs with small fracture networks.

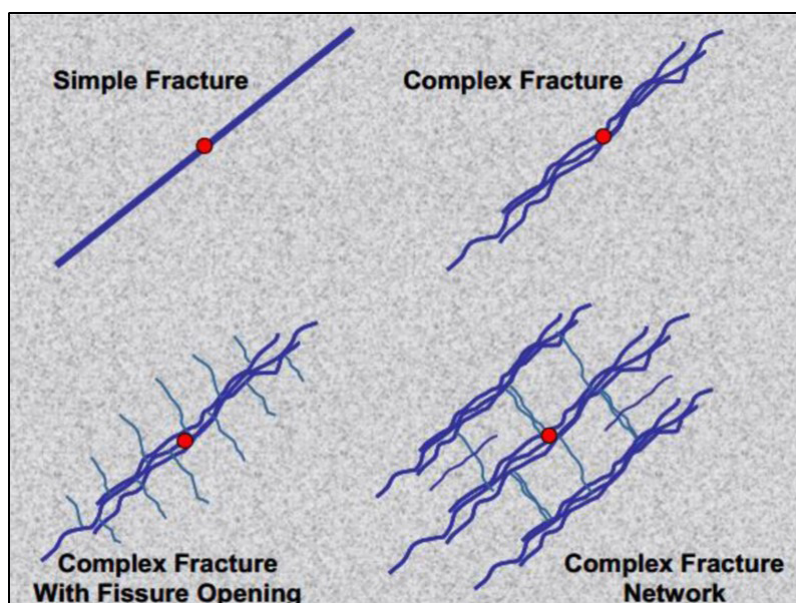


Fig. 9—Fracture complexity (Warpinski et al. 2009).

Alan Vardy's review on WH damping in surface pipelines highlights several key factors influencing this phenomenon, including unsteady friction on pipe walls, the viscoelastic behavior of plastic pipes, bubbly flows, gas flow in porous ducts, and fluid-structure interactions (Vardy 2023). The damping of the WH is also affected by fluid viscosity, pipe roughness, and the presence of bends and fittings in the pipelines. These elements add complexity to the interactions, playing a crucial role in absorbing and dissipating the energy of pressure waves, thereby mitigating the WH effects.

By drawing an analogy between WH in surface pipelines and hydraulic fracture jobs in the same horizontal well in unconventional reservoirs, it becomes clear that induced fracture complexity significantly contributes to the damping of WH effects observed after hydraulic fracturing stages. The activation of natural fractures during the fracturing stage acts as a dampener for the WH post-hydraulic fracture job. The complexity introduced by these natural fractures provides a mechanism for absorbing and reducing WH effects, particularly because the large fluid volumes used in hydraulic fracturing do not allow for gas segregation. All other parameters, such as the well, reservoir, perforation strategy, and hydraulic fracture design, remain constant. The key variable affecting the degree of WH dampening is the induced fracture complexity, with variations in the intensity of natural fractures within the fractured area. Complex fracture networks, as depicted in Fig. 10, serve to absorb and dissipate the energy of pressure waves, similar to how damping in a mechanical system reduces oscillations over time. This comprehensive analogy enhances our understanding of the dynamics of WH.

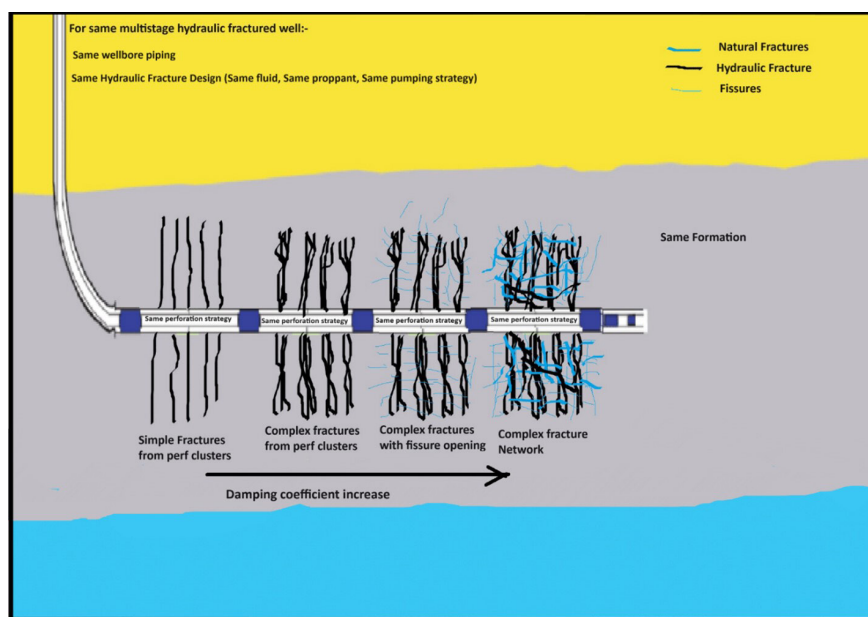


Fig. 10—Conceptual illustration of the effect of induced fracture complexity on damping coefficient.

In our study, the estimated damping coefficient using CWT is used to relatively estimate the induced fracture complexity along the wellbore. This approach can be validated by comparing it with the natural fracture interpretation for wells Boggess-3H, -5H, -9H, and

-13H from the Marcellus Shale Energy and Environment Laboratory project in the Marcellus Shale Formation and examples from the Middle Bakken formation. The interpretation of natural fractures from image logs provides a method to identify the natural fractures likely to be impacted by the fracturing fluid during hydraulic fracturing. This information can be utilized to infer the relative productivity of different fracture stages to calibrate the new technique for induced fracture complexity estimation.

One of the pads in the Marcellus Shale Energy and Environment Laboratory project is the Boggess pad, which consists of six horizontal Marcellus Shale wells, each extending over 10,000 ft, which were drilled from a single pad near the initial MIP pad. Each well is significantly stimulated with hydraulic fracture jobs, as seen in **Fig. 11**.

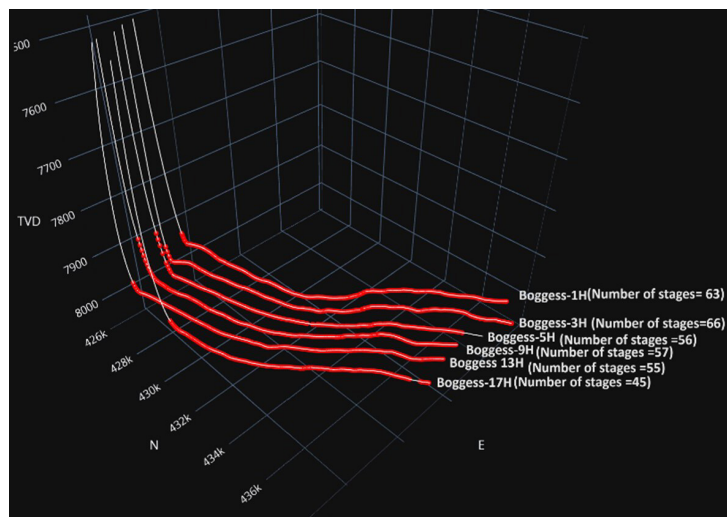


Fig. 11—Boggess pad wells in the Marcellus Shale Energy and Environment Laboratory project with their fracture stages. TVD = true vertical depth.

Post-hydraulic fracture pressure data often exhibit distinct WH signatures; however, some operations may not display these expected oscillating pressure wave patterns. The proposed model addresses this issue by simulating overdamped WH conditions with a higher value of the damping coefficient estimated using CWT. This added flexibility allows for the accurate representation of a wider range of WH scenarios, including those with unclear or absent WH signals. The model's robustness is demonstrated by its consistent ability to generate WH-like responses across multiple stages. The analysis highlights a correlation between overdamped WH signatures and an increase in the damping coefficient, which suggests a more complex fracture network. This complex propagation mode is characterized by poor communication between the wellbore and fractures, likely due to increased tortuosity. In the industry, the absence of WH signals is commonly used as an indicator of ineffective communication between the wellbore and fractures. Our model's assumptions are validated by the poor communication indicated by the increased damping coefficient (overdamping), which suggests induced fracture complexity.

Optimizing WH Model Parameters. In the pursuit of accurately modeling WH signals, nonlinear optimization algorithms are used to fine-tune the model parameters, ensuring alignment with observed data. This hyperparameter optimization process is essential in machine learning and statistical modeling, as it involves systematically adjusting the hyperparameters of a model to minimize the error between predicted and actual data. The optimization process aims to find the optimal set of hyperparameters that improve model performance, which is crucial for achieving accurate and reliable predictions.

The algorithms explored include differential evolution (DE), least squares, basin hopping, dual annealing, particle swarm optimization, Nelder-Mead, Rosenbrock, box (Powell technique), and constrained Rosenbrock (trust-constr). Each algorithm adopts a distinct strategy to explore the parameter space. DE, for example, is known for its robustness in handling high-dimensional, nonlinear, and nondifferentiable problems (Storn and Price 1997). The process begins with random initialization of a population of candidate solutions, followed by mutation and crossover operations to generate new solutions, and a selection step to retain the best candidates based on a defined fitness criterion. Despite its versatility, DE can experience slow convergence in highly complex problem spaces.

Least squares, another optimization technique, focuses on minimizing the sum of squared residuals, offering a straightforward approach for problems where the model's errors can be expressed as squared differences (Gauss and Stewart 1995). This method iteratively adjusts parameters to minimize the error, relying heavily on gradient information, and is particularly effective for smooth and well-behaved functions. However, it may be prone to getting stuck in local minima, especially when the initial parameter estimates are far from the optimal solution.

Basin hopping combines local optimization with stochastic perturbations, allowing the algorithm to escape local minima and explore the parameter space more thoroughly (Wales and Doye 2003). This method performs local optimization, perturbs the solution randomly, and accepts or rejects new solutions based on a probabilistic criterion, enhancing the likelihood of finding a global minimum. Dual annealing integrates the principles of simulated annealing with local optimization. It probabilistically explores the search space, accepting worse solutions with a decreasing probability to escape local minima and refine the solution iteratively (Kirkpatrick et al. 1983).

Particle swarm optimization is inspired by the social behavior of birds and fish. It uses a swarm of particles that explore the search space, each adjusting its position based on its own experience and that of its neighbors. Particle swarm optimization is particularly noted for its simplicity and ease of parallelization, making it effective for global optimization tasks, though it may require numerous iterations to converge to an optimal solution (Kennedy and Eberhart 1995).

Nelder-Mead operates on a simplex, iteratively refining it based on function evaluations at its vertices (Nelder and Mead 1965). This method is particularly useful for small-scale problems where gradient information is unavailable, though it can struggle with high-dimensional problems and may converge to local minima. Rosenbrock, a quasi-Newton method, approximates the Hessian matrix to

navigate efficiently toward the minimum, making it suitable for smooth, convex problems but less effective on nonconvex surfaces (Zhao 2021).

The box (Powell) technique, a derivative-free optimization method, is advantageous for problems with discontinuities or nondifferentiable functions. It explores the search space by adjusting step sizes based on function behavior but can be slow for large-scale problems and sensitive to the initial guess (Powell 1964). Constrained Rosenbrock (trust-constr) is a trust-region method that effectively manages both equality and inequality constraints, defining a region around the current solution where a model of the objective function is trusted to be accurate (Byrd et al. 2000).

The comparison of these algorithms underscores the importance of selecting the appropriate optimization strategy based on the problem's characteristics. Hyperparameter optimization, by systematically adjusting the hyperparameters, plays a critical role in enhancing model accuracy and reliability. DE is particularly recommended for tackling highly nonlinear and multimodal objective functions, while least squares is suitable for problems where errors can be minimized as squared residuals. Basin hopping and dual annealing are robust against local minima, offering a broad exploration of the solution space. The constrained Rosenbrock method excels in managing constraints, making it versatile for various optimization challenges. The choice of optimization algorithm, whether it be particle swarm optimization for its parallelization benefits or Nelder-Mead for simplicity in gradient-free scenarios, significantly impacts the efficiency and success of the modeling process.

All the previously described techniques were tried, and the best method was chosen for modeling WH; the best method was selected based on error from the real data-based Huber loss function. The Huber loss function is a robust and versatile error metric commonly used in regression problems, particularly when the data contains outliers. Unlike the mean squared error and mean absolute error (MAE), which treat all errors uniformly, the Huber loss function (Hastie et al. 2009) combines the advantages of both. It is defined as follows: For residuals smaller than a specified threshold δ , the loss behaves like mean squared error, providing a quadratic penalty that is sensitive to small errors. For residuals larger than δ , it transitions to a linear penalty similar to MAE, which reduces the influence of outliers. This characteristic blend ensures that the Huber loss is both robust to outliers and responsive to small deviations, making it a preferred choice in many practical applications. The mathematical formulation is shown in Eq. 12:

$$\begin{aligned} L_{\delta}(y, f(x)) &= \frac{1}{2} (y - f(x))^2 \text{ for } |y - f(x)| \leq \delta \\ L_{\delta}(y, f(x)) &= \delta (|y - f(x)| - \frac{1}{2} \delta) \text{ for } |y - f(x)| > \delta, \end{aligned} \quad (12)$$

where y represents the true value and $f(x)$ denotes the predicted value.

In this study, these optimization methods were systematically applied to optimize the model parameters. The effectiveness of each optimization technique was assessed using the Huber loss function, ensuring robustness against outliers. The parameters that resulted in the minimum Huber loss were considered optimal.

Results and Discussion

Utilizing the proposed model and applying CWT with a complex Morlet wavelet, we estimated the damping coefficient for different fracture stages along the lateral length in the Boggess-3H, -5H, -9H, and -13H wells. The damping coefficient can be plotted with reference to the middle of each stage depth, providing a well log for the damping coefficient distribution over the lateral length of the wells, as shown in Fig. 12a for Boggess-3H. In fracture interpretation from image logs for the four horizontal wells (Boggess-3H, -5H, -9H, and -13H), various types of fractures are classified based on their characteristics. Low-amplitude fractures are typically smaller and less prominent, indicating minimal displacement or disturbance within the rock formation. High-amplitude fractures are more significant, showing greater displacement or separation, often associated with higher fluid flow potential. Open fractures are those where the fracture walls are not in contact, allowing fluid movement through the openings. Faults represent significant structural breaks in the rock, usually with considerable displacement, and can act as major pathways for or barriers to fluid flow. The total count encompasses all these types, providing a comprehensive overview of the fracture network within the surveyed area. The analysis indicates well-developed natural fractures within the Marcellus formation, which are expected to influence stimulation efforts. The identified fractures were 1,732 in -3H, 1,475 in -5H, 1,362 in -9H, and 1,738 in -13H, with most being low amplitude. The fractures predominantly follow northeast/southwest orientation, with secondary sets trending sub-north-south and east-southeast/west-southwest. Fracture density is generally high but varies across and within wells, with organic-rich shale zones showing higher density, suggesting better potential for stimulation. Five minor faults were identified in Boggess-3H and one in Boggess-13H, with none detected in Boggess-5H and -9H. Structural dips across the wells were minimal, with mean dips ranging from 1.4° to 2°.

The presence of natural fractures is anticipated to facilitate the development of a more complex hydraulic fracture network, potentially increasing stimulated reservoir volume. Natural fracture density significantly influences the complexity of induced fractures during hydraulic fracturing operations. As hydraulic fractures propagate, their interaction with existing natural fractures can complicate the fracture network. The density and orientation of these natural fractures dictate the pathways of induced fractures, resulting in complex geometries that impact reservoir performance, as illustrated in Fig. 9. That is the main reason for introducing the damping coefficient as a quick cost-efficient tool for understanding the induced fracture complexity and natural fracture intensity along a horizontal well completed with the same design of multistage hydraulic fracturing as normally practiced.

Table 1 summarizes the fracture interpretation for the four horizontal wells (Boggess-3H, -5H, -9H, and -13H) along with the damping coefficient summation for all their fracture stages. Notably, the damping coefficient summation for each well strongly correlates with the number of open fractures detected along the lateral length, highlighting the role of open fractures in inducing fracture complexity.

The interpretation of natural fractures, derived from publicly available data and based on Blackriver Geoscience LLC's analysis for Northeast Natural Energy, was utilized to calibrate the distribution of the damping coefficient as a proxy for natural fracture intensity. This calibration is depicted in Figs. 12a and 12b for the Boggess-3H well. A comparison between the damping coefficient in Fig. 12a and the fracture density log at the top of Fig. 12b reveals a strong correlation (correlation coefficient = 0.68), suggesting that the trends in both logs are consistent. The damping coefficient trend effectively represents the average fracture density over the stage length, indicating induced fracture complexity.

In Fig. 12a, the damping coefficient exhibits significant variability across measured depths, with prominent spikes at approximately 13,300 ft, 14,000 ft, and 19,400 ft, indicating increased fracture activity in these intervals. Likewise, Fig. 12b presents the fracture density log, highlighting elevated values at similar depths, where the peaks closely align with those of the damping coefficient. This correlation implies that higher damping coefficients may indicate more complex fracture systems within these specific depth ranges.

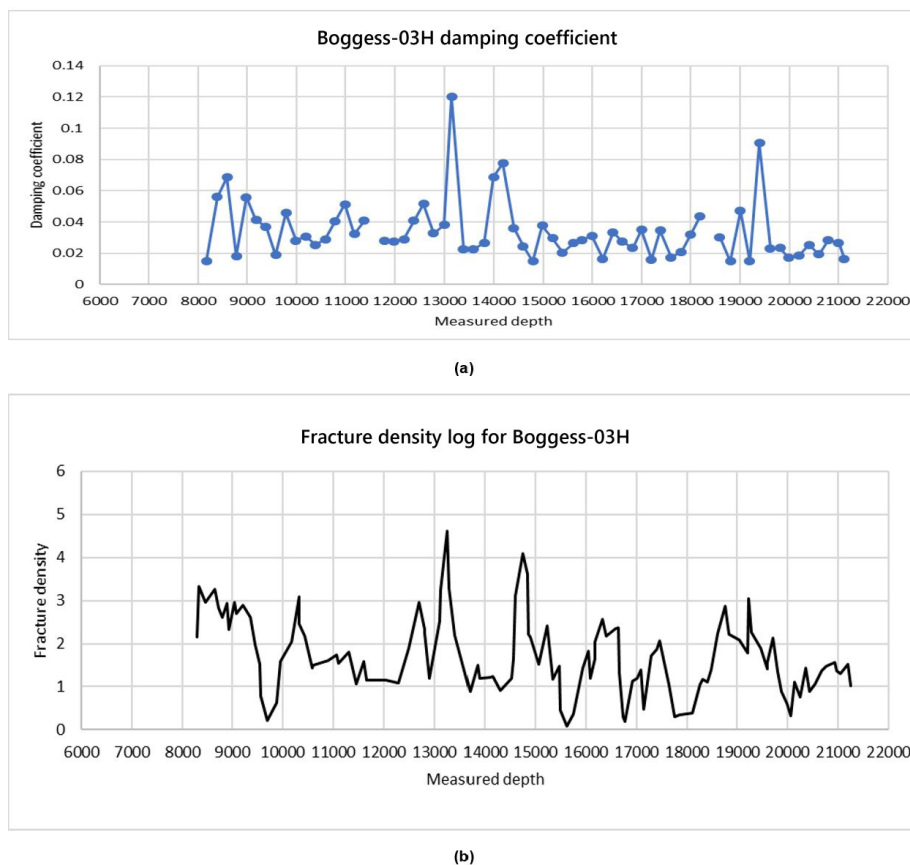


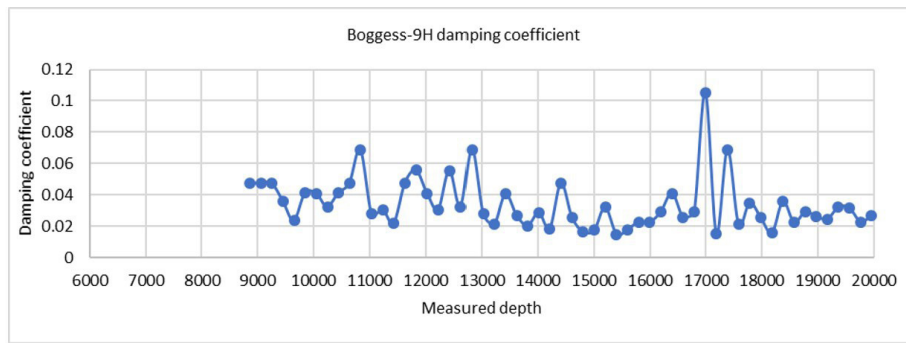
Fig. 12—(a) Damping coefficient across Boggess-3H fracture stages, referenced to midstage depth. (b) Fracture density along Boggess-3H wellbore [based on Blackriver Geoscience LLC interpretation prepared for Northeast Natural Energy and shared for public (digitized with moving average)].

	Summation of Damping Coefficient	Low-Amplitude Fractures	High-Amplitude Fractures	Open Fractures	Fault	Total
Boggess-3H	Invalid summation due to bad pump shutdown strategy in some stages	1,580	133	14	5	1,732
Boggess-5H	2.38	1,435	36	4	0	1,475
Boggess-9H	1.95	1,269	90	3	0	1,362
Boggess-13H	1.44	1,704	32	1	1	1,738

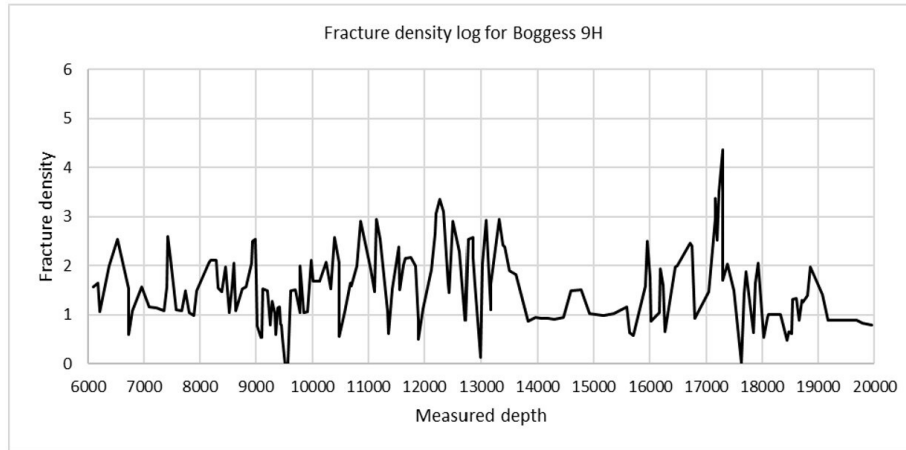
Table 1—Summary of fracture interpretation for the four horizontal wells (Boggess-3H, -5H, -9H, and -13H) with their damping coefficient summation for all their fracture stages.

A similar correlation is observed in **Figs. 13a and 13b** for the Boggess-9H well, with a correlation coefficient of 0.90. In **Fig. 13a**, the damping coefficient shows notable variations, particularly with peaks at around 10,800 ft, between 12,000 ft and 13,000 ft, and at approximately 17,000 ft. Correspondingly, **Fig. 13b** displays the fracture density log, revealing heightened values at these same depths, emphasizing spikes at 10,800 ft, between 12,000 ft and 13,000 ft, and at around 17,000 ft. This correlation suggests that regions with elevated damping coefficients are likely associated with more complex induced fracture systems, as indicated by the increased fracture density.

The relationship between the damping coefficient and fracture density, as observed in **Figs. 14 and 15** for the Boggess-3H and -9H wells, respectively, demonstrates a positive correlation where higher damping coefficients are generally associated with increased fracture density. The scatter shows some variability, with Boggess-9H displaying a wider range of fracture densities and damping coefficients compared with the more tightly clustered distribution in Boggess-3H.



(a)



(b)

Fig. 13—(a) The damping coefficient across Bogges-9H fracture stages, referenced to midstage depth. (b) Fracture density along Bogges-9H wellbore [based on Blackriver Geoscience LLC interpretation prepared for Northeast Natural Energy and shared for public (digitized with moving average)].

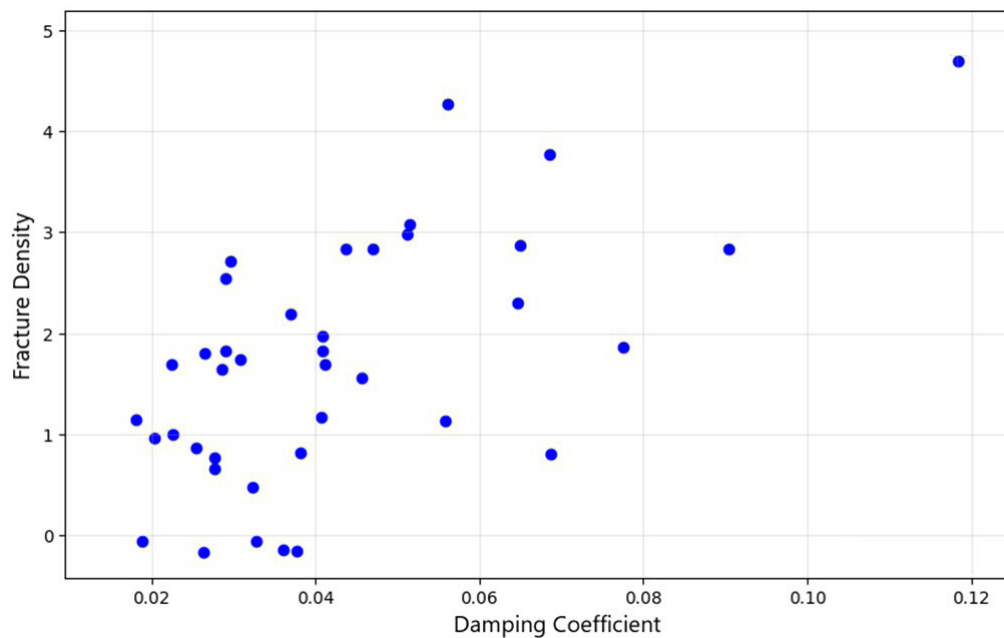


Fig. 14—Fracture density vs. damping coefficient for Bogges-3H.

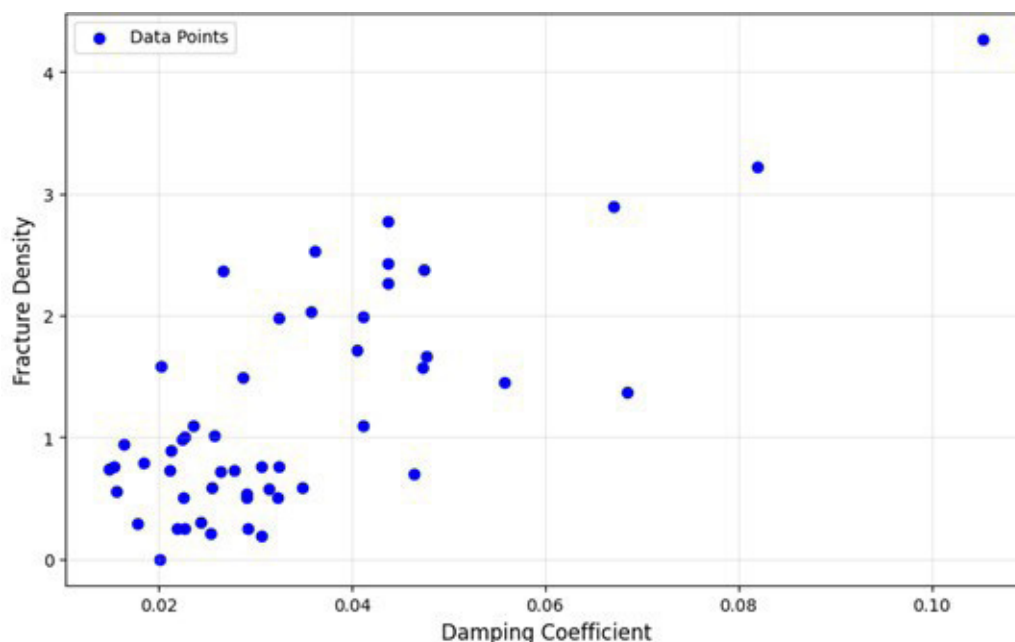


Fig. 15—Fracture density vs. damping coefficient for Bogges-9H.

After estimating the damping coefficient, the remaining parameters were determined through nonlinear optimization using various techniques. The nonlinear parameter optimization was selected as the WH propagation in the wellbore and fracture geometry systems in the nonlinear problem. The optimization method yielding the least error was selected as the best technique to accurately match the real WH data. **Figs. 16a, 16b, 16c, and 16d** present several examples of WH matches with the proposed model for randomly selected stages from the Bogges pad wells, respectively. These examples demonstrate the model's ability to accurately simulate various WH signals, including overdamped ones. The distribution of the best optimization methods used to match the real data was as follows: basinhopping at 58.68% and DE at 40.97%.

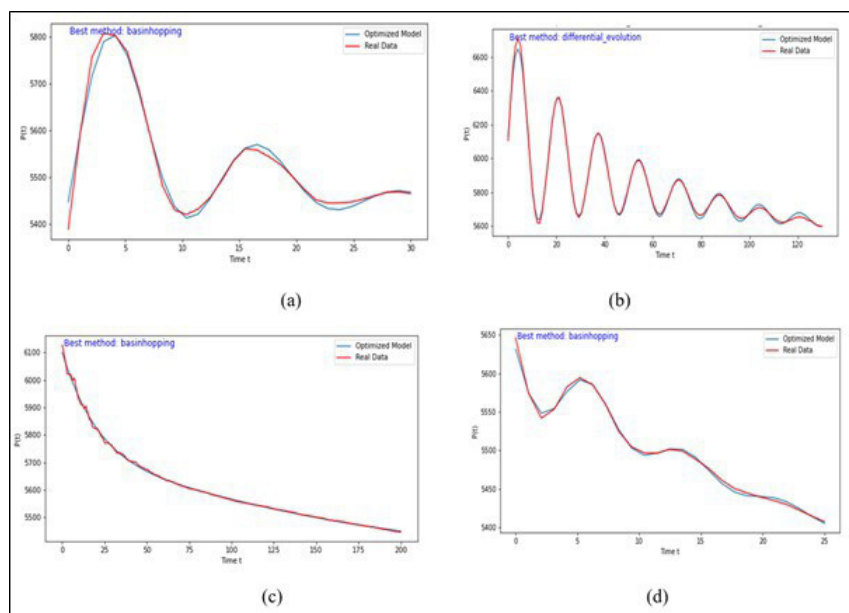


Fig. 16—Matching WH for hydraulic fracture job on (a) Bogges-1H Stage 30, (b) Bogges-9H Stage 2, (c) Bogges-3H Stage 66, and (d) Bogges-5H Stage 53.

The error analysis for the model is evaluated through key metrics: root mean squared error (RMSE), MAE, and R^2 . As shown in **Table A-1** in Appendix A, the error analysis of the model across various fracture stages reveals a mix of predictive accuracy and areas needing refinement. The RMSE ranges from 20.35 psi to 83.57 psi, suggesting variability in model performance, where lower RMSE indicates closer alignment with observed values, while higher RMSE points to greater discrepancies in certain stages. This variability could be due to unaddressed complexities or inconsistencies within the data set, requiring adjustments to model parameters or inputs for better accuracy. Similarly, the MAE fluctuates between 11.74 psi and 40.90 psi, reflecting the average magnitude of errors; stages with higher MAE may benefit from targeted calibration to reduce consistent deviations. The difference between RMSE and MAE across stages

suggests that some outliers might be disproportionately influencing the RMSE, given its sensitivity to larger errors. Despite these variations, the R^2 values remain above 0.86 across all stages, with some reaching up to 0.99, demonstrating the model's overall effectiveness. The parameters used to match the model have the distribution characteristics shown in **Fig. A-1** in Appendix A, where M , a , b , c , θ , and f exhibit distinct patterns of spread, symmetry, and concentration. Parameters such as b and c display broader distributions, indicating greater variability, while a and f show narrower, peaked distributions, reflecting strong consistency across the data set. M presents a uni-modal distribution with moderate variability, while θ demonstrates a relatively symmetric and clustered distribution. These distributions highlight the variability and stability of each parameter, offering valuable insights into their roles and sensitivity in the model-matching process.

The model aimed to replicate the WH pressure oscillations using a damped harmonic oscillator approach. However, its accuracy is contingent upon certain conditions being met. **Figs. 17a and 17b** illustrate a case where the model did not successfully match the observed data. Despite using the best optimization method (basinhopping), the model (blue line) diverged significantly from the real data (red line). The real data exhibited irregular pressure wave oscillations, suggesting the presence of noise and complex interactions that the model failed to capture.

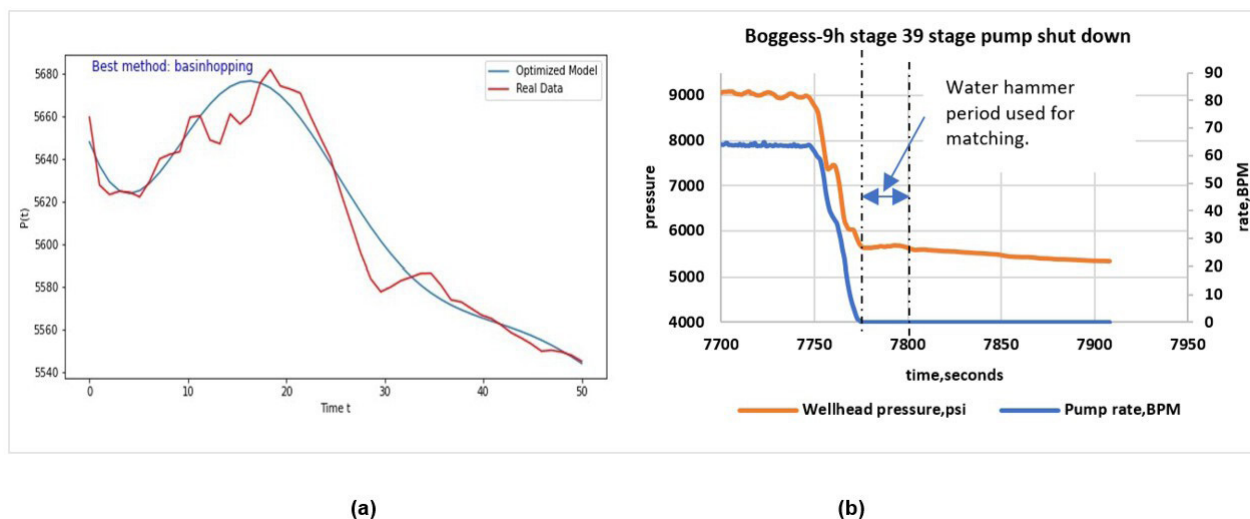


Fig. 17—(a) Example of model failure in matching a WH event and (b) pump shutdown strategy for same example.

This discrepancy highlights the model's limitations, particularly in scenarios where the WH event arises from a noninstantaneous pump shutdown. Gradual or stepped shutdowns produce multiple overlapping WH waves, leading to a complex superposition that the model struggles to simulate accurately. The irregularities and noise in the pressure data, as shown in **Fig. 17a**, exemplify these challenges. In this fracture stage, the model achieved an R^2 of 0.93, reflecting these complexities. The pump shutdown strategy for this stage, depicted in **Fig. 17b**, involved a 25-second, multistep process that generated multiple pressure waves throughout the system. To enhance the accuracy of WH modeling, it is crucial to allow the wave to propagate freely without interference from varying pumping rates, which requires an immediate shutdown—from maximum flow rate to zero. This approach would minimize extraneous noise, provide a clearer representation of the system's response, and facilitate better alignment between the model and real world data.

Conclusion

This study highlights the effectiveness of advanced signal processing techniques, specifically the CWT with a complex Morlet wavelet, in analyzing WH effects in hydraulic fracture systems. The main findings are as follows:

1. The CWT was utilized to estimate damping coefficients across various fracture stages, providing a detailed distribution log along the lateral lengths of wells. This revealed a strong correlation between damping coefficient distribution and natural fracture density, suggesting that damping coefficients are a reliable and cost-effective measure of induced fracture complexity and natural fracture intensity.
2. WH effects were simulated using a damped harmonic oscillator model, which demonstrated a range of behaviors, from high-amplitude oscillations to overdamped scenarios. The model's accuracy depended on an immediate pump shutdown; gradual shutdowns complicated the superposition of pressure waves.
3. Nonlinear optimization techniques, particularly basinhopping, effectively fine-tuned model parameters to match real WH data, highlighting their utility in optimizing model accuracy.

These findings underscore the potential of CWT in enhancing fracture characterization and wellbore dynamics analysis, offering a detailed and cost-effective approach to assessing fracture complexity and damping coefficients, with significant implications for future research in the field.

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Appendix A

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boggess_13h10	2,747.134	52.41311	30.02985	0.911727
Boggess_13h11	1,986.166	44.56642	30.96154	0.924764
Boggess_13h12	2,647.457	51.45345	24.57504	0.893245
Boggess_13h13	6,984.743	83.57478	40.89794	0.865211
Boggess_13h14	413.9334	20.34535	11.74465	0.992617
Boggess_13h15	851.883	29.18704	14.80079	0.98889
Boggess_13h16	1,918.259	43.79793	20.79823	0.974506
Boggess_13h17	1,844.142	42.94348	27.29633	0.920825
Boggess_13h18	183.0082	13.52805	8.528957	0.993338
Boggess_13h19	2,567.627	50.67175	20.82362	0.913142
Boggess_13h1	1,0198.99	100.99	49.541	0.871061
Boggess_13h20	2,339.08	48.36404	27.86397	0.920096
Boggess_13h21	662.066	25.73064	13.06849	0.977869
Boggess_13h22	3,414.127	58.43053	27.49719	0.891632
Boggess_13h23	3,298.376	57.43149	32.62158	0.914361
Boggess_13h24	5,062.136	71.14869	33.06208	0.793938
Boggess_13h25	798.7759	28.26262	16.34788	0.984171
Boggess_13h26	1,565.226	39.56293	22.92555	0.935852
Boggess_13h27	1,301.135	36.07126	26.49276	0.942651
Boggess_13h28	3,035.392	55.09439	24.92915	0.945666
Boggess_13h29	5,656.671	75.21085	34.33058	0.85124
Boggess_13h2	429.5433	20.72543	12.26666	0.993889
Boggess_13h30	5,901.261	76.81966	34.51378	0.857951
Boggess_13h31	353.0291	18.78907	9.873966	0.994797
Boggess_13h32	603.0067	24.55619	11.68248	0.959224
Boggess_13h33	117.0937	10.82098	5.939439	0.986569
Boggess_13h34	1,330.935	36.48198	17.01128	0.942227
Boggess_13h35	93.84574	9.687401	7.153994	0.988933
Boggess_13h36	734.6149	27.10378	13.15452	0.949184
Boggess_13h37	2,067.012	45.4644	20.33983	0.946621
Boggess_13h38	825.4012	28.7298	16.19121	0.977558
Boggess_13h39	5,492.167	74.10916	35.64221	0.870918
Boggess_13h3	3,811.857	61.74024	35.42949	0.931366
Boggess_13h40	1,319.27	36.32176	19.88729	0.910854
Boggess_13h41	1,295.115	35.9877	20.29219	0.936914
Boggess_13h42	1,032.472	32.1321	24.56917	0.915469
Boggess_13h43	1,153.347	33.96097	18.23276	0.949448
Boggess_13h44	80.33006	8.962704	6.074559	0.96066
Boggess_13h45	621.2208	24.9243	11.04655	0.903664
Boggess_13h46	732.7612	27.06956	16.453	0.9518
Boggess_13h47	256.2021	16.00631	7.960747	0.990566

Table A-1—Error metrics for WH model.

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boggess_13h48	814.6916	28.5428	16.89614	0.973996
Boggess_13h4	301.5629	17.36557	8.957794	0.967519
Boggess_13h50	1,335.537	36.545	22.07375	0.945954
Boggess_13h51	124.3064	11.14928	6.19706	0.996145
Boggess_13h52	330.8214	18.1885	9.937418	0.981904
Boggess_13h53	2,075.078	45.55302	23.40811	0.950204
Boggess_13h54	242.7014	15.57888	9.425203	0.988184
Boggess_13h55	516.7235	22.73155	14.89103	0.933863
Boggess_13h5	281.8591	16.78866	13.35167	0.971162
Boggess_13h6	904.5671	30.07602	17.53141	0.98599
Boggess_13h7	283.7177	16.84392	9.625774	0.97975
Boggess_13h8	1,332.511	36.50357	17.37417	0.922264
Boggess_13h9	6,244.087	79.01954	38.06815	0.84731
Boggess_17h10	3,165.981	56.26705	28.67378	0.894252
Boggess_17h11	4,457.954	66.76791	34.80382	0.87627
Boggess_17h12	7,932.305	89.06349	38.28572	0.895897
Boggess_17h13	939.7694	30.65566	13.57164	0.957034
Boggess_17h14	6,155.515	78.45709	36.32483	0.887994
Boggess_17h15	5,931.971	77.01929	38.0105	0.865005
Boggess_17h16	3,272.491	57.20569	30.58475	0.994178
Boggess_17h17	6,872.439	82.90018	39.05203	0.85865
Boggess_17h18	1,032.936	32.13932	19.16028	0.945379
Boggess_17h19	1,212.875	34.82635	14.67545	0.96618
Boggess_17h1	165.8788	12.8794	9.045986	0.998119
Boggess_17h21	211.1994	14.5327	11.58894	0.993348
Boggess_17h22	307.029	17.52224	9.419	0.993528
Boggess_17h23	752.2863	27.42784	17.28589	0.981615
Boggess_17h24	192.385	13.87029	8.623046	0.996607
Boggess_17h25	378.4662	19.45421	12.37326	0.991751
Boggess_17h26	357.6015	18.91036	14.6955	0.984267
Boggess_17h27	3,660.897	60.50535	33.25864	0.92109
Boggess_17h28	2,492.945	49.9294	21.1794	0.952518
Boggess_17h29	5,423.881	73.647	58.37727	0.892038
Boggess_17h2	3,720.181	60.99328	28.58912	0.934581
Boggess_17h31	385.3168	19.62949	16.18953	0.987069
Boggess_17h32	9.620829	3.101746	1.93058	0.999267
Boggess_17h33	3,575.163	59.79267	46.36953	0.861269
Boggess_17h34	1,690.488	41.11554	33.19156	0.963939
Boggess_17h35	4,390.573	66.2614	52.47796	0.843685
Boggess_17h36	580.2587	24.08856	17.779	0.709192
Boggess_17h37	1,148.212	33.88528	21.76345	0.82875
Boggess_17h38	336.7777	18.3515	9.224406	0.984716
Boggess_17h39	865.7402	29.42346	21.20387	0.929763
Boggess_17h3	413.65	20.33839	12.53072	0.987719
Boggess_17h40	453.2756	21.29027	10.26716	0.974586
Boggess_17h41	2,707.419	52.03287	35.92882	0.89845
Boggess_17h42	3.255558	1.804316	1.210205	0.998133
Boggess_17h43	11.0669	3.326695	2.238949	0.995452

Table A-1 (continued)—Error metrics for WH model.

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boggess_17h44	600.6444	24.50805	18.04972	0.96957
Boggess_17h45	313.1431	17.69585	5.947729	0.977831
Boggess_17h4	383.163	19.57455	11.25298	0.973369
Boggess_17h5	7,337.496	85.65919	69.63985	0.949508
Boggess_17h6	4,503.889	67.11102	54.65711	0.962435
Boggess_17h7	27.93608	5.285459	3.844821	0.996839
Boggess_17h8	123.4212	11.10951	8.87687	0.996189
Boggess_17h9	1,512.635	38.89261	19.02326	0.977352
Boggess_1h10	77.506	8.803749	4.473987	0.898307
Boggess_1h11	4,968.455	70.48727	31.39342	0.863035
Boggess_1h12	109.0524	10.44282	6.064175	0.992886
Boggess_1h13	541.4432	23.26893	16.66201	0.993966
Boggess_1h14	361.1997	19.00525	12.83558	0.922991
Boggess_1h15	826.7324	28.75295	14.46908	0.972817
Boggess_1h16	2,900.709	53.85823	44.29483	0.965367
Boggess_1h17	1,362.475	36.91171	23.5873	0.927983
Boggess_1h19	7,047.407	83.94884	38.43192	0.845004
Boggess_1h20	259.582	16.11155	7.056068	0.97468
Boggess_1h21	224.9656	14.99885	11.94183	0.99566
Boggess_1h22	5,699.856	75.49739	32.72603	0.872528
Boggess_1h23	294.7584	17.16853	10.53585	0.98574
Boggess_1h24	2,308.698	48.04891	34.81235	0.919121
Boggess_1h25	609.1279	24.68052	13.76118	0.974808
Boggess_1h26	969.5388	31.13742	24.55875	0.976945
Boggess_1h27	1,427.062	37.77648	22.36747	0.968066
Boggess_1h28	2,628.629	51.27016	21.99465	0.87664
Boggess_1h29	2,441.108	49.40757	28.14303	0.924522
Boggess_1h2	8,183.554	90.463	46.58653	0.807825
Boggess_1h30	212.1874	14.56665	10.31108	0.984351
Boggess_1h31	801.1477	28.30455	17.16972	0.981346
Boggess_1h32	209.8556	14.48639	7.803637	0.992123
Boggess_1h33	963.2542	31.03634	15.86836	0.968997
Boggess_1h34	1,200.943	34.65462	27.43303	0.983701
Boggess_1h35	3,743.682	61.18564	27.36228	0.885972
Boggess_1h36	1,088.944	32.99915	16.3886	0.974286
Boggess_1h37	534.3705	23.11646	14.42125	0.931242
Boggess_1h38	2,058.318	45.36869	19.6174	0.940947
Boggess_1h3	227.0009	15.06655	10.22053	0.986452
Boggess_1h40	2,149.662	46.36445	21.34911	0.921571
Boggess_1h41	1,602.222	40.02777	19.11642	0.958952
Boggess_1h42	75.7311	8.702362	5.77058	0.994294
Boggess_1h43	11.54488	3.397775	2.498267	0.998331
Boggess_1h45	1.156057	1.075201	0.858696	0.999971
Boggess_1h46	34.20823	5.84878	3.852712	0.996146
Boggess_1h47	44.27557	6.653989	3.981469	0.975881
Boggess_1h48	393.3649	19.83343	14.34124	0.964621
Boggess_1h49	48.98305	6.998789	5.118435	0.990292
Boggess_1h4	243.5924	15.60745	12.37222	0.988323

Table A-1 (continued)—Error metrics for WH model.

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boggess_1h50	227.7242	15.09053	8.300445	0.99041
Boggess_1h51	11.98731	3.462269	1.833887	0.990957
Boggess_1h52	0.866449	0.930832	0.750612	0.980637
Boggess_1h53	5.92388	2.433902	1.658685	0.998073
Boggess_1h54	144.0402	12.00168	7.959675	0.918104
Boggess_1h55	2.329716	1.526341	1.247171	0.994987
Boggess_1h56	46.27912	6.802876	5.405159	0.997445
Boggess_1h57	22.31157	4.723513	3.317656	0.825642
Boggess_1h58	3.669283	1.915537	1.437223	0.998385
Boggess_1h59	115.9043	10.76588	7.600303	0.989282
Boggess_1h5	90.77431	9.527555	5.403384	0.987285
Boggess_1h60	602.4751	24.54537	17.03234	0.917624
Boggess_1h61	17.76033	4.214301	2.711112	0.994252
Boggess_1h62	46.01363	6.783335	4.669849	0.998301
Boggess_1h63	1.23368	1.110711	0.781664	0.999796
Boggess_1h6	10448.07	102.2158	45.71747	0.85886
Boggess_1h7	645.3444	25.40363	15.12916	0.991873
Boggess_1h8	113.3988	10.64889	8.595703	0.99149
Boggess_1h9	287.6729	16.96092	11.18885	0.993041
Boggess_3h10	7.603436	2.757433	2.187658	0.999562
Boggess_3h11	5,995.636	77.43149	34.96465	0.854622
Boggess_3h12	1,152.375	33.94666	27.31616	0.989125
Boggess_3h13	6,497.153	80.60492	33.9019	0.801498
Boggess_3h14	1,544.439	39.29935	23.27751	0.949151
Boggess_3h16	964.0706	31.04949	22.81988	0.986872
Boggess_3h17	1,753.79	41.87827	22.1574	0.928275
Boggess_3h18	997.8957	31.58949	13.06375	0.956335
Boggess_3h19	10,941.49	104.6016	46.86089	0.848487
Boggess_3h1	6,983.178	83.56541	46.8077	0.940891
Boggess_3h20	605.5121	24.60716	18.6096	0.999517
Boggess_3h21	5,566.737	74.61057	33.30689	0.817169
Boggess_3h22	202.244	14.22125	10.59256	0.996628
Boggess_3h23	521.0518	22.82656	11.62016	0.960892
Boggess_3h24	714.1723	26.724	14.46221	0.984091
Boggess_3h25	361.6947	19.01827	11.46493	0.993817
Boggess_3h26	3,103.931	55.71293	25.80004	0.914191
Boggess_3h27	709.1993	26.6308	15.42214	0.900078
Boggess_3h28	180.2725	13.42656	8.848132	0.993235
Boggess_3h29	5,569.344	74.62804	41.38611	0.828026
Boggess_3h2	4,375.203	66.14532	29.58283	0.896111
Boggess_3h30	1,101.882	33.19461	15.80758	0.949379
Boggess_3h31	5,021.562	70.86298	40.78969	0.922912
Boggess_3h32	792.5692	28.15261	19.27748	0.966665
Boggess_3h33	3,393.516	58.25389	22.7423	0.903553
Boggess_3h34	525.1163	22.91542	13.69539	0.987486
Boggess_3h35	403.833	20.0956	15.11123	0.993467
Boggess_3h36	195.6392	13.98711	9.249272	0.968259
Boggess_3h37	1,204.096	34.70009	27.46019	0.969864

Table A-1 (continued)—Error metrics for WH model.

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boguess_3h38	1,487	38.56164	18.16957	0.952058
Boguess_3h39	331.3529	18.2031	9.885666	0.984192
Boguess_3h3	8,619.656	92.84211	49.04228	0.820201
Boguess_3h40	156.5285	12.51114	7.378757	0.971465
Boguess_3h41	15,250.76	123.494	89.76477	0.845695
Boguess_3h42	46.28138	6.803042	4.195808	0.996345
Boguess_3h43	349.7642	18.70198	14.22247	0.984445
Boguess_3h44	1,825.397	42.72467	32.75137	0.956601
Boguess_3h45	280.4914	16.74788	11.61411	0.985945
Boguess_3h46	717.6732	26.78942	14.23121	0.961779
Boguess_3h47	201.389	14.19116	7.952442	0.985453
Boguess_3h48	2,309.986	48.06231	26.67386	0.863477
Boguess_3h4	4,421.498	66.49434	41.15631	0.885598
Boguess_3h50	84.72366	9.204545	4.940179	0.992801
Boguess_3h51	1,089.047	33.00071	17.37391	0.955778
Boguess_3h52	218.8271	14.79281	8.952021	0.982975
Boguess_3h53	955.0426	30.90376	19.15753	0.929448
Boguess_3h54	224.7584	14.99195	10.19341	0.974708
Boguess_3h55	1,001.036	31.63915	18.51961	0.955832
Boguess_3h56	408.1278	20.20217	10.44745	0.978562
Boguess_3h57	324.477	18.01324	8.571847	0.99137
Boguess_3h58	18466.1	135.89	75.64506	0.841428
Boguess_3h59	960.1271	30.98592	12.02674	0.976283
Boguess_3h5	426.449	20.65064	11.73273	0.980189
Boguess_3h60	3,245.324	56.96774	38.10789	0.865723
Boguess_3h61	23,811.33	154.3092	80.30544	0.790146
Boguess_3h62	13.02188	3.608584	2.661264	0.997625
Boguess_3h63	572.164	23.91995	7.609754	0.948309
Boguess_3h65	147.5744	12.14802	6.089756	0.987504
Boguess_3h66	38.11856	6.174023	3.289839	0.998252
Boguess_3h6	5,697.966	75.48487	41.60367	0.89432
Boguess_3h7	3,494.369	59.11319	25.95772	0.933349
Boguess_3h8	3,386.421	58.19296	28.15677	0.971431
Boguess_3h9	7,428.958	86.19141	44.24176	0.905153
Boguess_5h11	163.5522	12.78875	10.23769	0.988565
Boguess_5h12	33.64806	5.800695	4.653716	0.996797
Boguess_5h13	64.61652	8.03844	5.973161	0.995954
Boguess_5h15	83.08569	9.115135	7.344789	0.997652
Boguess_5h16	322.3507	17.95413	10.3612	0.983884
Boguess_5h17	173.7154	13.18011	7.561896	0.991331
Boguess_5h18	182.9866	13.52725	9.39631	0.975058
Boguess_5h19	0.022776	0.150918	0.119333	0.99998
Boguess_5h20	1046.641	32.35183	20.39261	0.849657
Boguess_5h21	34.90873	5.908361	4.123563	0.989505
Boguess_5h22	900.8016	30.01336	14.71696	0.945144
Boguess_5h23	683.1702	26.13753	14.98356	0.922516
Boguess_5h24	436.9521	20.9034	14.98231	0.998319
Boguess_5h25	958.2248	30.95521	15.8923	0.973232

Table A-1 (continued)—Error metrics for WH model.

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boguess_5h26	60.8536	7.800872	5.495668	0.999401
Boguess_5h27	652.4385	25.54288	12.33335	0.952963
Boguess_5h28	1,571.614	39.64358	19.57335	0.868342
Boguess_5h30	632.6863	25.15326	15.79246	0.942414
Boguess_5h32	140.3654	11.84759	9.044473	0.910062
Boguess_5h33	1,678.524	40.96979	21.80156	0.905528
Boguess_5h34	1,458.269	38.18729	19.80967	0.916803
Boguess_5h35	173.7149	13.1801	8.188384	0.991684
Boguess_5h36	456.3998	21.36352	12.3333	0.949393
Boguess_5h37	321.1926	17.92185	8.446764	0.963103
Boguess_5h38	203.8097	14.27619	10.62354	0.979703
Boguess_5h39	597.6657	24.4472	14.40019	0.978479
Boguess_5h40	416.0851	20.39816	11.00236	0.94723
Boguess_5h41	24.70382	4.970293	3.387627	0.992026
Boguess_5h42	209.8858	14.48744	10.84013	0.99153
Boguess_5h43	94.76353	9.734656	7.368676	0.968265
Boguess_5h44	330.3006	18.17417	13.67932	0.975355
Boguess_5h45	1,298.201	36.03056	16.98644	0.935132
Boguess_5h46	267.5681	16.35751	12.38316	0.9529
Boguess_5h47	182.7697	13.51924	8.86073	0.972986
Boguess_5h48	697.8911	26.41763	15.21792	0.973707
Boguess_5h49	119.3229	10.9235	5.405393	0.978968
Boguess_5h50	355.2929	18.84921	10.99499	0.791093
Boguess_5h51	331.4508	18.20579	11.22942	0.94489
Boguess_5h52	288.9206	16.99766	9.887282	0.980205
Boguess_5h53	17.88151	4.228654	2.839555	0.995549
Boguess_5h54	10.24947	3.201479	2.105512	0.995687
Boguess_5h55	47.02947	6.857803	4.63493	0.989663
Boguess_5h56	0.902137	0.949809	0.689616	0.999869
Boguess_9h10	10,634.57	103.124	47.88016	0.821609
Boguess_9h11	885.9539	29.76498	14.64865	0.972892
Boguess_9h12	278.9055	16.70046	13.60971	0.99569
Boguess_9h13	2,813.029	53.038	28.36441	0.950717
Boguess_9h14	104.0382	10.19991	6.272698	0.986792
Boguess_9h15	2,521.93	50.21882	22.66475	0.950873
Boguess_9h16	196.22	14.00785	10.34785	0.957812
Boguess_9h17	2,299.942	47.95771	21.91757	0.922847
Boguess_9h18	1,403.699	37.46597	19.44447	0.983603
Boguess_9h19	461.5939	21.48474	12.78994	0.980258
Boguess_9h1	287.3452	16.95126	13.44778	0.996827
Boguess_9h20	1,091.189	33.03315	18.92264	0.939835
Boguess_9h21	192.7629	13.88391	9.596183	0.986007
Boguess_9h22	3,278.737	57.26026	27.08424	0.886369
Boguess_9h23	2,362.278	48.60327	25.30909	0.971881
Boguess_9h24	353.7636	18.8086	10.04984	0.982336
Boguess_9h25	889.9797	29.83253	18.19858	0.951249
Boguess_9h26	2,986.912	54.65265	24.35675	0.91339
Boguess_9h27	1,536.268	39.19525	21.66803	0.954419

Table A-1 (continued)—Error metrics for WH model.

Fracture Stage	Mean Squared Error	RMSE	MAE	R ²
Boguess_9h28	661.096	25.71179	12.85331	0.988036
Boguess_9h29	281.1275	16.76686	8.081116	0.979788
Boguess_9h2	249.1974	15.78599	10.86253	0.995891
Boguess_9h30	3,663.314	60.52532	29.4017	0.869888
Boguess_9h31	1,130.712	33.62606	20.6482	0.97058
Boguess_9h32	346.2356	18.60741	8.92144	0.96638
Boguess_9h33	3,679.68	60.66037	33.3643	0.931437
Boguess_9h34	1,300.095	36.05683	23.53752	0.948
Boguess_9h35	835.9575	28.91293	15.35057	0.979363
Boguess_9h36	1,787.121	42.27435	24.22847	0.946391
Boguess_9h38	1,332.028	36.49697	22.56417	0.974984
Boguess_9h39	124.9668	11.17885	7.641435	0.932505
Boguess_9h3	43.16779	6.57022	4.886632	0.998594
Boguess_9h40	1,757.135	41.9182	21.55357	0.916241
Boguess_9h41	1,204.583	34.7071	19.96563	0.955406
Boguess_9h42	687.3694	26.21773	15.7292	0.940453
Boguess_9h43	1,086.693	32.96502	19.32482	0.976004
Boguess_9h44	1,300.129	36.0573	18.43912	0.949909
Boguess_9h45	127.4809	11.29075	6.570688	0.995852
Boguess_9h46	810.3278	28.46626	14.94322	0.957689
Boguess_9h47	381.5509	19.53333	12.17119	0.894234
Boguess_9h48	48.10115	6.935499	5.433564	0.99739
Boguess_9h49	183.8955	13.56081	8.735316	0.993455
Boguess_9h4	717.1203	26.7791	17.8626	0.980007
Boguess_9h50	572.0085	23.9167	13.6857	0.980227
Boguess_9h51	93.25232	9.656724	5.466973	0.99573
Boguess_9h52	585.9284	24.20596	14.53009	0.975562
Boguess_9h53	1,881.078	43.37139	21.64188	0.928528
Boguess_9h54	131.7895	11.47996	7.633153	0.991973
Boguess_9h55	781.8857	27.96222	16.65712	0.964329
Boguess_9h56	88.82004	9.424439	5.991743	0.993494
Boguess_9h57	190.0641	13.78637	9.197897	0.967219
Boguess_9h5	370.435	19.24669	12.46361	0.994961
Boguess_9h6	311.1844	17.64042	12.7481	0.996985
Boguess_9h7	644.3344	25.38374	17.04316	0.987129
Boguess_9h8	2,655.455	51.5311	29.75578	0.951134
Boguess_9h9	3,501.409	59.1727	34.46575	0.879393

Table A-1 (continued)—Error metrics for WH model.

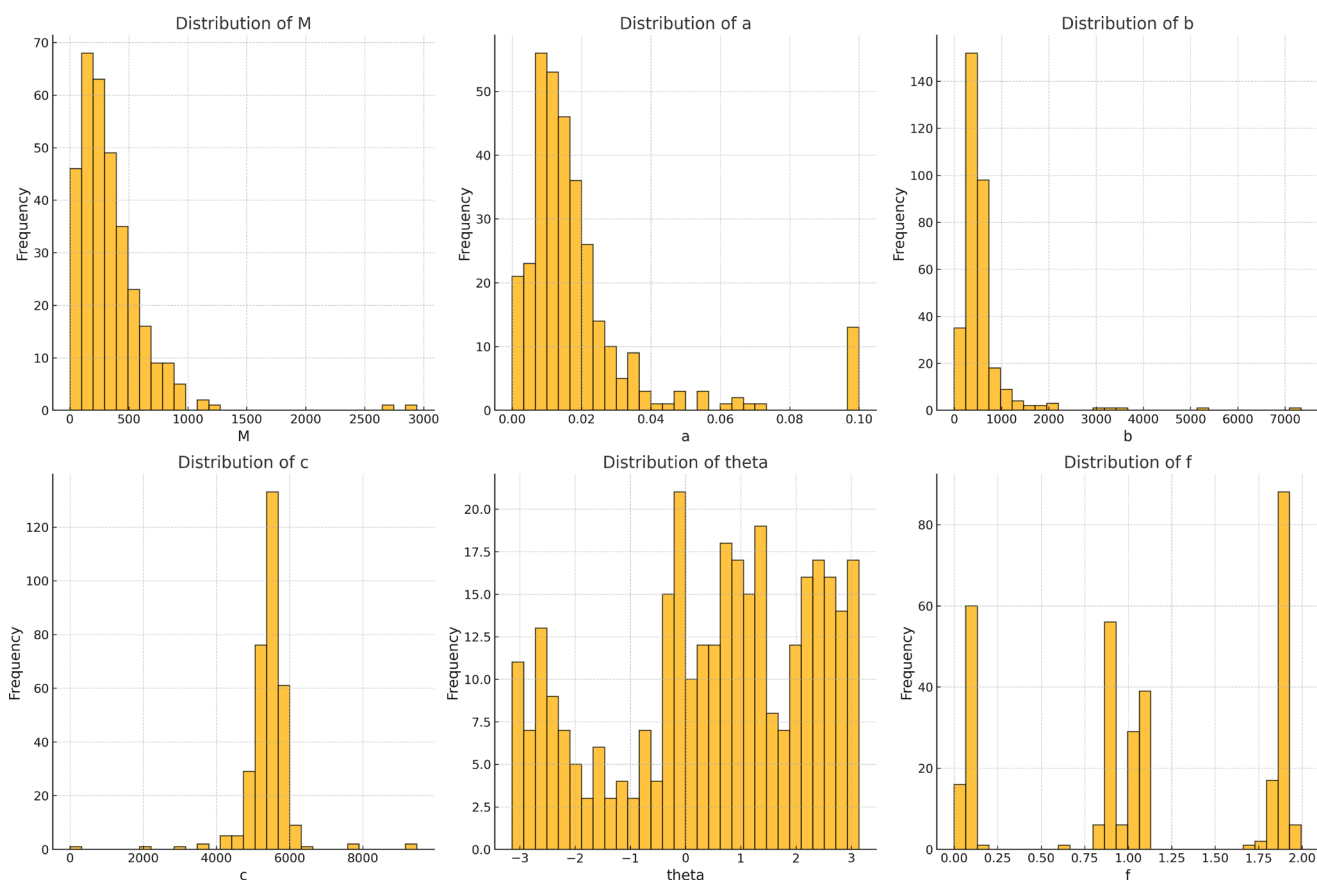


Fig. A-1—Distribution of parameters used to match WH model.